Supplementary Material

SignMuon, MuonSign, and the Role of Error Feedback

A Appendix

A.1 Preliminaries: LMO, Muon, and generalized smoothness

This subsection states in full the definitions that Section 3 uses in abbreviated form.

A *Linear Minimization Oracle* (LMO) is an operator $A(\cdot)$ that provides the solution to a linear minimization problem over a feasible set. In particular, the constraints can be defined via the unit ball of a given norm:

$$A(\mathbf{X}) = \arg\min_{\mathbf{Y}\in\{\mathbf{Y}\in\mathcal{S}\mid \|\mathbf{Y}\|\le 1\}} \langle \mathbf{X}, \mathbf{Y}\rangle, \tag{1}$$

where $\mathbf{X} \in \mathbb{R}^{m\times n}$ is the input matrix, $\mathcal{S} \subset \mathbb{R}^{m\times n}$ is a convex set, and $\langle \mathbf{X}, \mathbf{Y}\rangle = \sum_{i,j} X_{ij}Y_{ij}$ is the standard matrix inner product (Pethick et al. 2025a). Such an operator selects the direction of steepest descent with respect to the specified norm. Its analytical and practical role is discussed in detail in the literature on norm-constrained LMO optimizers (Pethick et al. 2025a; Kovalev 2025; Riabinin et al. 2025; Bernstein and Newhouse 2024; Bernstein 2025), and the catalogue of usable geometries keeps growing (Cesista 2025; Kravatskiy et al. 2025).

Muon is the instance in which each matrix layer carries the RMS→RMS operator norm, equal to $\sqrt{n/m}\,\|\cdot\|_{2\to 2}$ on $\mathbb{R}^{m\times n}$, so that the oracle uses the matrix structure of the gradient and the update direction is obtained by orthogonalizing the gradient matrix (Bernstein and Newhouse 2024). The two norms differ by a constant per layer, which we carry in the step size rather than in the geometry (Appendix A.17); every statement below is therefore in the spectral norm. Let $\mathbf{M} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$ be the singular value decomposition (SVD) of the matrix $\mathbf{M} \in \mathbb{R}^{m\times n}$, where $\mathbf{U} \in \mathbb{R}^{m\times r}$ and $\mathbf{V} \in \mathbb{R}^{n\times r}$ are the orthonormal matrices of singular vectors, $\mathbf{\Sigma} \in \mathbb{R}^{r\times r}$ is the diagonal matrix of singular values, and $r = \operatorname{rank}(\mathbf{M})$. Then the LMO direction is $A(\mathbf{M}) = -\mathbf{U}\mathbf{V}^\top$: Muon selects an orthonormal update direction corresponding to the solution of the linear minimization problem induced by the spectral norm geometry. Such orthogonalization produces more isotropic updates and prevents a small number of dominant singular directions from controlling the optimization process, which is particularly beneficial for training DNN (Bernstein and Newhouse 2024; Shah et al. 2025). In the original Muon implementation the computation of $\mathbf{U}_t\mathbf{V}_t^\top$ is approximated using Newton-Schulz iterations and related polynomial schemes, which enable efficient matrix orthogonalization without an explicit SVD computation (Amsel et al. 2025; Li and Hong 2025; Liu et al. 2025); Algorithm A1 gives the procedure and the coefficients we use.

Finally, Corollary 2 requires Assumption 2 only in its weaker layer-wise (L^0, L^1) form, which replaces the constant L_i^g by $L_i^{0,g} + L_i^{1,g}\|\nabla_i g(\mathbf{X})\|_*$ (Riabinin et al. 2025; Pethick et al. 2025b): for every layer i and all $\mathbf{X}, \mathbf{Y}$,

$$\|\nabla_i g(\mathbf{X}) - \nabla_i g(\mathbf{Y})\|_* \le \left(L_i^{0,g} + L_i^{1,g}\|\nabla_i g(\mathbf{X})\|_*\right)\|\mathbf{X}_i - \mathbf{Y}_i\|_{2\to 2}, \tag{2}$$

again with one pair of constants per layer for $g = f$ and per layer and client for $g = f_j$.

A.2 Width-one blocks: vector parameters

The setup of Section 3 asks each block to be a matrix, and $\min(m_i, n_i) = 1$ is permitted: a bias, a normalization gain, or any other one-dimensional parameter is the block $\mathbb{R}^{1\times n_i}$ or $\mathbb{R}^{m_i\times 1}$. Gluon states the same product space and leaves the norm on each block arbitrary, so it too admits them without comment (Riabinin et al. 2025); Scion is explicit, giving biases the RMS norm with oracle $\mathbf{b}/\|\mathbf{b}\|_{\mathrm{RMS}}$ (Pethick et al. 2025a). Fixing the spectral norm on every block recovers nearly that: on a width-one block the rank is one, the spectral, nuclear and Euclidean norms coincide, and $\operatorname{polar}(\mathbf{g}) = \mathbf{g}/\|\mathbf{g}\|_2$ for $\mathbf{g} \ne \mathbf{0}$. Assumptions 1 and 2 then read unchanged, and the norm-equivalence constant of Corollary 1 improves to $\bar\rho_i = \sqrt{r_i} = 1$.

What such a block cannot do is diverge. Substituting $\operatorname{polar}(\mathbf{g}) = \mathbf{g}/\|\mathbf{g}\|_2$ into (6) gives, on $\mathbb{R}^{1\times n}$,

$$\operatorname{sign}(\operatorname{polar}(\mathbf{g})) = \operatorname{sign}(\mathbf{g}), \qquad \operatorname{polar}(\operatorname{sign}(\mathbf{g})) = \tfrac{1}{\sqrt{n}}\operatorname{sign}(\mathbf{g}),$$

and the two-sided placement returns $\operatorname{sign}(\mathbf{g})$ as well: all three are SignSGD up to a positive constant the step size absorbs, and $\langle \mathbf{g}, \mathbf{s}(\mathbf{g})\rangle$ is $\|\mathbf{g}\|_1$ or $\|\mathbf{g}\|_1/\sqrt{n}$, positive either way. No instance of Theorems 1–3 can live on a width-one block, which agrees with the two minimality analyses below: the $O(2)$ computation of Appendix A.4, and the sign-pattern enumeration of Appendix A.5, which finds no mismatch at $2\times n$ for $n \le 12$. Composing a sign with the oracle is harmless exactly where the oracle has nothing to orthogonalize; the smallest ascent instance we exhibit is 4×4.

The unit-gain multipliers of Appendix A.17 need no special case either: $\lambda = \sqrt{\text{fan-out}}/\|\mathbf{s}\|_F$ returns $(1,\, 1/\sqrt{n})$ for the LMO and SIGN families on $\mathbb{R}^{1\times n}$, and $(\sqrt{m},\, 1)$ on $\mathbb{R}^{m\times 1}$.

Which parameters reach the methods. Little of this is load-bearing in our experiments, since the sign methods are applied where Muon is. Centralized and federated runs give the rule to parameters of two or more dimensions other than the classifier head, and route biases, BatchNorm scales and that head to AdamW as the auxiliary group, whose bandwidth cost Appendix A.14 counts. On nanoGPT we keep record #40's grouping unchanged: one-dimensional scalars, the embeddings and the head go to its distributed Adam, the hidden matrices and the two gate weights to the method under test. One gate is width-one — the 1×12 `smear_gate` — so the method is exactly SignSGD there. Our implementation returns one-dimensional tensors from the oracle unchanged rather than ℓ_2-normalized, which differs from polar by a positive scale and so alters neither the sign nor the descent inner product.

A.3 Divergence on linear objectives: reduction and momentum

The reduction quoted in Section 4 collapses each of the three methods to a single scalar inequality and removes momentum from the discussion entirely.

Proposition 1 (Reduction on linear objectives) *Run any of the three methods* (6) *on the linear objective* (8) *with* $\mathbf{G} \neq \mathbf{0}$*, from an arbitrary* $\mathbf{X}_0$*, with any momentum coefficient* $\mu \in [0, 1)$ *under either the Standard or the Nesterov rule. Then* $\tilde{\mathbf{M}}_t = \gamma_t \mathbf{G}$ *with* $\gamma_t > 0$*, the update direction is the constant matrix* $\mathbf{s}(\mathbf{G})$ *obtained by substituting* $\mathbf{G}$ *for* $\tilde{\mathbf{M}}_t$ *in* (6)*, and*

$$f(\mathbf{X}_t) - f(\mathbf{X}_{t-1}) = -\eta_t \big\langle \mathbf{G},\ \mathbf{s}(\mathbf{G}) \big\rangle.$$

In particular, if $\langle \mathbf{G}, \mathbf{s}(\mathbf{G}) \rangle < 0$ *then* f *strictly increases at every iteration for any* $\eta_t > 0$*, and* $f(\mathbf{X}_t) \to +\infty$ *whenever* $\sum_t \eta_t = \infty$ *(for instance, for any constant step size).*

Proof of Proposition 1. On the linear objective (8) the gradient is globally constant, $\mathbf{G}_t = \mathbf{G}$, so the momentum buffer of (5) is $\mathbf{M}_t = (1-\mu) \sum_{i=0}^{t-1} \mu^i \, \mathbf{G} = (1-\mu^t)\, \mathbf{G}$. Both momentum rules then return a positive multiple of $\mathbf{G}$,

$$\tilde{\mathbf{M}}_t = \gamma_t \, \mathbf{G}, \qquad \gamma_t = \begin{cases} 1 - \mu^t, & \text{(Standard)}, \\ 1 - \mu^{t+1}, & \text{(Nesterov)}, \end{cases} \tag{3}$$

both positive for $t \geq 1$ because $\mu \in [0, 1)$; the Nesterov case is $(1-\mu)\mathbf{G} + \mu(1-\mu^t)\mathbf{G}$. (Under the heavy-ball convention every γ_t is multiplied by $1/(1-\mu)$, which changes nothing below.) The elementwise $\operatorname{sign}(\cdot)$ and the Muon LMO $\operatorname{polar}(\cdot)$ are each invariant under multiplication by a positive scalar, so evaluating (6) at $\tilde{\mathbf{M}}_t = \gamma_t \mathbf{G}$ returns the same matrix as evaluating it at $\mathbf{G}$; that is, $\mathbf{s}_t = \mathbf{s}(\mathbf{G})$ for every t, independently of μ and of the momentum variant. Hence $f(\mathbf{X}_t) - f(\mathbf{X}_{t-1}) = \langle \mathbf{G}, \mathbf{X}_t - \mathbf{X}_{t-1} \rangle = -\eta_t \langle \mathbf{G}, \mathbf{s}(\mathbf{G}) \rangle$, which is (9). If $\langle \mathbf{G}, \mathbf{s}(\mathbf{G}) \rangle < 0$, then $f(\mathbf{X}_t) = f(\mathbf{X}_0) - \langle \mathbf{G}, \mathbf{s}(\mathbf{G}) \rangle \sum_{i=1}^t \eta_i$ increases strictly at every step and diverges to $+\infty$ whenever $\sum_t \eta_t = \infty$. ■

By Proposition 1, each divergence theorem reduces to exhibiting a single gradient $\mathbf{G}$ with $\langle \mathbf{G}, \mathbf{s}(\mathbf{G}) \rangle < 0$; the three proofs below accomplish exactly this, and momentum requires no further comment.

A.4 Proof of Theorem 1 (Divergence of SignMuon)

Theorem 1 (Divergence of SignMuon) *There is a matrix* $\mathbf{G} \in \mathbb{R}^{4\times 4}$ *with* $\big\langle \mathbf{G}, \operatorname{sign}(\operatorname{polar}(\mathbf{G})) \big\rangle = -\frac{42468}{103} < 0$*. Hence SignMuon ascends on* (8)*:* f *strictly increases at every iteration for all* $\eta_t > 0$*, all* $\mu \in [0, 1)$*, and both momentum variants.*

For SignMuon $\mathbf{s}(\mathbf{G}) = \operatorname{sign}(\operatorname{polar}(\mathbf{G}))$, so by (9) it suffices to construct $\mathbf{G} \in \mathbb{R}^{4\times 4}$ with $\langle \mathbf{G}, \operatorname{sign}(\operatorname{polar}(\mathbf{G})) \rangle < 0$. The size is not arbitrary. Writing the polar decomposition $\mathbf{G} = \mathbf{Q}\mathbf{H}$ with $\mathbf{H} \succeq 0$, and using that $\operatorname{polar}(\mathbf{G}) = \mathbf{Q}$ and that $\mathbf{H}$ is symmetric,

$$\langle \mathbf{G}, \operatorname{sign}(\operatorname{polar}(\mathbf{G})) \rangle = \big\langle \mathbf{H}, \operatorname{sym}(\mathbf{Q}^\top \operatorname{sign}(\mathbf{Q})) \big\rangle, \tag{4}$$

so a counterexample with polar factor $\mathbf{Q}$ exists precisely when $\lambda_{\min}\big(\operatorname{sym}(\mathbf{Q}^\top \operatorname{sign}(\mathbf{Q}))\big) < 0$, a question concerning $O(n)$ alone. For $n = 2$ this settles the matter. Every $\mathbf{Q} \in O(2)$ is $\left(\begin{smallmatrix} c & -s \\ s & c \end{smallmatrix}\right)$ or $\left(\begin{smallmatrix} c & s \\ s & -c \end{smallmatrix}\right)$ with $c^2 + s^2 = 1$; whenever $cs \neq 0$, both cases give $\mathbf{Q}^\top \operatorname{sign}(\mathbf{Q}) = (|c| + |s|)\mathbf{I} + (\text{antisymmetric})$, whose symmetric part is positive definite, so that $\langle \mathbf{G}, \operatorname{sign}(\operatorname{polar}(\mathbf{G})) \rangle = (|c| + |s|) \operatorname{tr} \mathbf{H} > 0$. When $cs = 0$ the randomized signs of the two vanishing entries enter, and $\operatorname{sym}(\mathbf{Q}^\top \operatorname{sign}(\mathbf{Q}))$ has eigenvalues $1 \pm |\varrho|$ with $\varrho \in \{-1, 0, 1\}$ the mean of those two signs, hence remains positive semidefinite. In either case $\lambda_{\min} \geq 0$ and SignMuon cannot ascend. At $n = 3$ several independent searches over $O(3)$ produced no negative eigenvalue; at $n = 4$ they do, and the $\mathbf{O}$ below is one such instance (its spectrum is $(-0.4175, 1.5825, 2, 2)$, so it is not even the most extreme).

We construct $\mathbf{G} \in \mathbb{R}^{4\times 4}$ by defining a specific orthogonal matrix $\mathbf{O}$ and a specific rank-1 principal component $\mathbf{u}_1 \mathbf{v}_1^\top$. Let the orthogonal matrix $\mathbf{O}$ be given by the following exact rational numbers:

$$\mathbf{O} = \frac{1}{103} \begin{pmatrix} 101 & 20 & 2 & -2 \\ -20 & 97 & 20 & -20 \\ -2 & 20 & 2 & 101 \\ -2 & 20 & -101 & -2 \end{pmatrix}. \tag{5}$$

Because no entry is zero, its element-wise sign matrix $\mathbf{S} = \operatorname{sign}(\mathbf{O})$ is uniquely defined. Now, let $\mathbf{u}_1$ and $\mathbf{v}_1$ be the following exact unit vectors:

$$\mathbf{u}_1 = \frac{1}{\sqrt{309}}\begin{pmatrix}10\\-3\\10\\10\end{pmatrix}, \quad \mathbf{v}_1 = \frac{1}{\sqrt{309}}\begin{pmatrix}10\\3\\-10\\10\end{pmatrix}. \tag{6}$$

One can easily verify that $\mathbf{O}\mathbf{v}_1 = \mathbf{u}_1$, meaning $\mathbf{u}_1$ and $\mathbf{v}_1$ act perfectly as left and right singular vectors for this orthogonal space. The crucial feature of this geometry is that the Frobenius inner product between this rank-1 component and the sign matrix $\mathbf{S}$ yields an exact, strictly negative fraction:

$$\langle \mathbf{u}_1\mathbf{v}_1^\top, \mathbf{S}\rangle = \mathbf{u}_1^\top \mathbf{S}\mathbf{v}_1 = -\frac{43}{103}. \tag{7}$$

We construct the gradient matrix $\mathbf{G}$ by assigning a large singular value ($\sigma_1 = 1001$) to this pathological component and a singular value of 1 to the rest of the orthogonal space:

$$\mathbf{G} \approx \begin{pmatrix}324.6 & 97.3 & -323.6 & 323.6\\ -97.3 & -28.2 & 97.3 & -97.3\\ 323.6 & 97.3 & -323.6 & 324.6\\ 323.6 & 97.3 & -324.6 & 323.6\end{pmatrix}. \tag{8}$$

Indeed, since $\mathbf{G} = 1000\,\mathbf{u}_1\mathbf{v}_1^\top + \mathbf{O}$ and $\mathbf{O}\mathbf{v}_1 = \mathbf{u}_1$,

$$\begin{aligned}\mathbf{G}\mathbf{v}_1 &= 1000\,\mathbf{u}_1(\mathbf{v}_1^\top\mathbf{v}_1) + \mathbf{O}\mathbf{v}_1\\ &= 1001\,\mathbf{u}_1,\end{aligned}$$

so $(\mathbf{u}_1, \mathbf{v}_1)$ is a singular pair of $\mathbf{G}$ with $\sigma_1 = 1001$. For any $\mathbf{w} \perp \mathbf{v}_1$ we have $\mathbf{G}\mathbf{w} = \mathbf{O}\mathbf{w}$; as $\mathbf{O}$ is orthogonal and $\mathbf{O}\mathbf{v}_1 = \mathbf{u}_1$, the restriction $\mathbf{O}|_{\mathbf{v}_1^\perp}$ is an isometry onto $\mathbf{u}_1^\perp$, hence $\sigma_2 = \sigma_3 = \sigma_4 = 1$. Moreover, with $\mathbf{u}_k = \mathbf{O}\mathbf{v}_k$ for $k \geq 2$,

$$\textstyle\sum_{k\geq 2} \mathbf{u}_k\mathbf{v}_k^\top = \mathbf{O}\big(\mathbf{I} - \mathbf{v}_1\mathbf{v}_1^\top\big) = \mathbf{O} - \mathbf{u}_1\mathbf{v}_1^\top,$$

so $\mathbf{U}_G\mathbf{V}_G^\top = \mathbf{u}_1\mathbf{v}_1^\top + (\mathbf{O} - \mathbf{u}_1\mathbf{v}_1^\top) = \mathbf{O}$. All four singular values are positive, so $\mathbf{G}$ is invertible and $\operatorname{polar}(\mathbf{G}) = \mathbf{O}$ is its unique polar factor; the same holds at the 5×5 instance of Theorems 2–3, so none of the three placements is refuted at an argument where the oracle has a choice to make.

Substituting the resulting matrix $\mathbf{G}$ into the descent condition yields:

$$\langle \mathbf{G}, \mathbf{S}\rangle = \langle 1000\,\mathbf{u}_1\mathbf{v}_1^\top + \mathbf{O}, \mathbf{S}\rangle = 1000\langle \mathbf{u}_1\mathbf{v}_1^\top, \mathbf{S}\rangle + \langle \mathbf{O}, \mathbf{S}\rangle. \tag{9}$$

The inner product $\langle \mathbf{O}, \mathbf{S}\rangle$ is equivalent to the L_1 norm (sum of absolute values) of $\mathbf{O}$, which equals exactly $\frac{532}{103}$. Therefore:

$$\langle \mathbf{G}, \mathbf{S}\rangle = 1000\left(-\frac{43}{103}\right) + \frac{532}{103} = -\frac{42468}{103} \approx -412.31. \tag{10}$$

Here $\mathbf{S} = \operatorname{sign}(\mathbf{O}) = \operatorname{sign}(\operatorname{polar}(\mathbf{G})) = \mathbf{s}(\mathbf{G})$, so $\langle \mathbf{G}, \mathbf{s}(\mathbf{G})\rangle = -\frac{42468}{103} < 0$. By Proposition 1, SignMuon strictly ascends, $f(\mathbf{X}_t) - f(\mathbf{X}_{t-1}) = \frac{42468}{103}\,\eta_t > 0$ at every iteration, for every $\eta_t > 0$, every $\mu \in [0, 1)$, and both momentum variants; $f(\mathbf{X}_t) \to +\infty$ under any non-summable step size. ■

A.5 Proof of Theorem 2 (Divergence of MuonUSign)

Theorem 2 (Divergence of MuonUSign) *There is a matrix* $\mathbf{G} \in \mathbb{R}^{5\times 5}$ *with* $\big\langle \mathbf{G}, \operatorname{polar}(\operatorname{sign}(\mathbf{G}))\big\rangle \approx -13.89 < 0$. *Hence MuonUSign ascends on* (8) *for all* $\eta_t > 0$, *all* $\mu \in [0, 1)$, *and both momentum variants.*

MuonUSign (Algorithm A4) applies the sign *before* the LMO, so $\mathbf{s}(\mathbf{G}) = \operatorname{polar}(\operatorname{sign}(\mathbf{G}))$. By (9) it suffices to construct $\mathbf{G} \in \mathbb{R}^{5\times 5}$ with $\langle \mathbf{G}, \operatorname{polar}(\operatorname{sign}(\mathbf{G}))\rangle < 0$.

The size is again not arbitrary, and here the question reduces to *sign patterns*. Put $\mathbf{S} = \operatorname{sign}(\mathbf{G})$ and $\mathbf{D} = \operatorname{polar}(\mathbf{S})$. Under the randomized convention $\mathbf{S} \in \{\pm 1\}^{m\times n}$ throughout, and the entrywise identity $G_{ij} = |G_{ij}|\, S_{ij}$ holds without exception, both sides vanishing wherever $G_{ij} = 0$. Consequently

$$\big\langle \mathbf{G}, \operatorname{polar}(\operatorname{sign}(\mathbf{G}))\big\rangle = \textstyle\sum_{i,j} |G_{ij}|\, S_{ij} D_{ij}, \tag{11}$$

and likewise $\langle \mathbf{G}, \operatorname{sign}(\mathbf{D})\rangle = \sum_{i,j} |G_{ij}|\, S_{ij} \operatorname{sign}(D_{ij})$ for the MuonSign step of Theorem 3. An ascent instance therefore exists at a given shape precisely when some sign pattern $\mathbf{S}$ there possesses a *mismatched* entry, $S_{ij}D_{ij} < 0$: sufficiency by inflating $|G_{ij}|$ at that entry, necessity because every summand in (11) is otherwise nonnegative. The existence of a mismatch is invariant under row and column sign flips, since $\operatorname{polar}(\mathbf{P}_1\mathbf{S}\mathbf{P}_2) = \mathbf{P}_1 \operatorname{polar}(\mathbf{S})\mathbf{P}_2$ for ± 1 diagonal $\mathbf{P}_1, \mathbf{P}_2$ leaves each product $S_{ij}D_{ij}$

unchanged, so enumerating one representative per class settles the question outright. The enumeration returns: *no* pattern at 4×4, at $3 \times n$ ($n \leq 9$) or at $2 \times n$ ($n \leq 12$) possesses a mismatch. Neither MuonUSign nor MuonSign can therefore ascend on any linear objective at those shapes whatever. The first instances appear at 4×5 (720 of the 2^{12} classes) and, among square shapes, at 5×5, where the $\mathbf{S}$ below attains the deepest mismatch any 5×5 sign matrix admits, $|D_{4,2}| = 1/\sqrt{17}$. (Had we retained $\operatorname{sign}(0) = 0$, gradients with exact zeros would admit instances from 3×4 onward; the convention eliminates them, together with their dependence on an entry being exactly zero.)

Fix the full-rank sign matrix $\mathbf{S} \in \{-1, 1\}^{5\times 5}$,

$$\mathbf{S} = \begin{pmatrix} -1 & -1 & 1 & 1 & 1 \\ -1 & -1 & 1 & -1 & -1 \\ 1 & -1 & 1 & 1 & -1 \\ 1 & 1 & -1 & -1 & 1 \\ 1 & 1 & 1 & -1 & 1 \end{pmatrix}, \tag{12}$$

and let $\mathbf{D} = \operatorname{polar}(\mathbf{S})$ be its (unique, since $\mathbf{S}$ is full rank) Muon LMO direction. A direct computation gives $D_{4,2} = -1/\sqrt{17} \approx -0.2425 < 0$ while $\operatorname{sign}(D_{i,j}) = S_{i,j}$ at all 24 other entries; equivalently, $\operatorname{sign}(\mathbf{D})$ and $\mathbf{S}$ disagree at exactly the single entry $(4, 2)$, where $S_{4,2} = +1$. We exploit this lone mismatch. Define

$$\mathbf{G} = \epsilon\,\mathbf{S} + (M - \epsilon)\,\mathbf{e}_4\mathbf{e}_2^\top, \qquad \epsilon > 0,\ M > \epsilon, \tag{13}$$

so that $G_{4,2} = M > 0$ and $G_{i,j} = \epsilon S_{i,j}$ otherwise; hence $\operatorname{sign}(\mathbf{G}) = \mathbf{S}$ and $\operatorname{polar}(\operatorname{sign}(\mathbf{G})) = \mathbf{D}$ for every $M > 0$. The descent inner product splits as

$$\langle \mathbf{G}, \mathbf{D} \rangle = \epsilon \sum_{(i,j)\neq(4,2)} S_{i,j} D_{i,j} \;+\; M\,D_{4,2} = \epsilon\,C + M\,D_{4,2}, \tag{14}$$

where $C := \sum_{(i,j)\neq(4,2)} |D_{i,j}| = 10.366 > 0$ is fixed (every such entry agrees in sign with $\mathbf{S}$), while $M\,D_{4,2} = -M/\sqrt{17} \to -\infty$. Thus $\langle \mathbf{G}, \mathbf{D} \rangle < 0$ for any $M > \sqrt{17}\,C\epsilon \approx 42.7\,\epsilon$; with $\epsilon = 1,\ M = 100$ one obtains $\langle \mathbf{G}, \mathbf{D} \rangle = -13.89$, for the exact polar factor that the theorem is stated over. By Proposition 1, MuonUSign strictly ascends on $f(\mathbf{X}) = \langle \mathbf{G}, \mathbf{X} \rangle$ for every $\eta_t > 0$, every $\mu \in [0, 1)$, and both momentum variants; $f(\mathbf{X}_t) \to +\infty$ under any non-summable step size. ■

A.6 Proof of Theorem 3 (Divergence of MuonSign)

Theorem 3 (Divergence of MuonSign) *For the* same *matrix* $\mathbf{G} \in \mathbb{R}^{5\times 5}$ *as in Theorem 2,* $\big\langle \mathbf{G}, \operatorname{sign}(\operatorname{polar}(\operatorname{sign}(\mathbf{G}))) \big\rangle = -76 < 0$. *Hence MuonSign ascends on* (8) *for all* $\eta_t > 0$, *all* $\mu \in [0, 1)$, *and both momentum variants.*

MuonSign (Algorithm A5) signs the LMO output as well, so $\mathbf{s}(\mathbf{G}) = \operatorname{sign}(\operatorname{polar}(\operatorname{sign}(\mathbf{G}))) = \operatorname{sign}(\mathbf{D})$. We reuse the *same* $\mathbf{S}$ and $\mathbf{G}$ of (12)–(13): since $\operatorname{sign}(\mathbf{G}) = \mathbf{S}$, the bidirectional step is the constant matrix $\operatorname{sign}(\mathbf{D})$, which agrees with $\mathbf{S}$ at all 24 entries except $(4, 2)$, where $\operatorname{sign}(D_{4,2}) = -1 = -S_{4,2}$. Using $S_{i,j}^2 = 1$ everywhere,

$$\begin{aligned} \langle \mathbf{G}, \operatorname{sign}(\mathbf{D}) \rangle &= \epsilon \sum_{(i,j)\neq(4,2)} S_{i,j} \operatorname{sign}(D_{i,j}) + M\ \operatorname{sign}(D_{4,2}) \\ &= 24\,\epsilon - M. \end{aligned} \tag{15}$$

This is negative for every $M > 24\epsilon$; with $\epsilon = 1,\ M = 100$ it equals exactly -76. By Proposition 1, MuonSign strictly ascends on $f(\mathbf{X}) = \langle \mathbf{G}, \mathbf{X} \rangle$ for every $\eta_t > 0$, every $\mu \in [0, 1)$, and both momentum variants; $f(\mathbf{X}_t) \to +\infty$ under any non-summable step size. In particular, the *same* 5×5 linear instance defeats both the uplink-only (MuonUSign) and the bidirectional (MuonSign) placements. ■

A.7 Comparison with the convergence claim for SignMuon

Concurrently, Mishra, Trivedi, and Kumar (2026) introduced the same sign-after-LMO method under the same name *SignMuon*, with update direction $\mathbf{S}_t = \operatorname{sign}(\operatorname{polar}(\mathbf{M}_t))$ rescaled to $\mathbf{D}_t = \mathbf{S}_t/\sqrt{mn}$; this coincides with the SignMuon step of Section 4 after absorbing $1/\sqrt{mn}$ into η. For this method they establish an $\mathcal{O}(1/\sqrt{T})$ stationarity guarantee. We show that this guarantee does not contradict Theorem 1: the finalized rate is proved for a different update direction, and for $\mathbf{D}_t$ their bound is uninformative on the instance of Theorem 1.

Their analysis controls the stationarity proxy $\mathcal{G}_T = \frac{1}{T}\sum_t \mathbb{E}[\|\mathbf{g}_t\|_1/\sqrt{mn}]$, where $\mathbf{g}_t = \nabla f(\mathbf{X}_t)$, through the per-entry sign-error probabilities $q_{ij,t} = \Pr(\mathbf{S}_{t,ij} \neq \operatorname{sign}(\mathbf{g}_{t,ij}) \mid \mathbf{X}_t)$, in two steps. First, for an arbitrary sign oracle $\mathbf{S}_t$ they derive

$$\begin{aligned} \mathcal{G}_T \leq{} & \frac{f(\mathbf{X}_0) - f^*}{\eta T} + \frac{L_*}{2}\,\eta \\ & + \underbrace{\frac{2}{T}\sum_t \mathbb{E}\left[\frac{1}{\sqrt{mn}} \sum_{i,j} |\mathbf{g}_{t,ij}|\, q_{ij,t}\right]}_{=:\,R_T}. \end{aligned} \tag{16}$$

Inequality (16) applies in particular to $\mathbf{S}_t = \mathrm{sign}(\mathrm{polar}(\mathbf{M}_t))$, but with the residual R_T left unbounded. Second, the rate is obtained by estimating R_T, and this estimate is derived *only* for the gradient-sign oracle $\mathbf{S}_t = \mathrm{sign}(\widetilde{\mathbf{G}}_t)$, where $\widetilde{\mathbf{G}}_t$ is an unbiased stochastic gradient with coordinatewise variance $\mathrm{Var}(\widetilde{\mathbf{G}}_{t,ij} \mid \mathbf{X}_t) \leq \sigma_{ij}^2/n_b$ and n_b the mini-batch size. In that case a Markov–Jensen argument yields $|\mathbf{g}_{t,ij}|\, q_{ij,t} \leq \sigma_{ij}/\sqrt{n_b}$, hence $R_T = \mathcal{O}(\|\sigma\|_1/\sqrt{mn\, n_b})$ with $\|\sigma\|_1 = \sum_{i,j}\sigma_{ij}$, which vanishes as $n_b \to \infty$.

This estimate bounds $q_{ij,t}$ by the noise level $\sigma_{ij}/\sqrt{n_b}$ and is therefore specific to $\mathrm{sign}(\widetilde{\mathbf{G}}_t)$, whose only source of sign error is gradient noise. It does not extend to the polar-sign oracle: even with exact gradients ($\sigma_{ij} = 0$) the latter satisfies $q_{ij,t} = \mathbf{1}[\mathrm{sign}(\mathrm{polar}(\mathbf{g}_t))_{ij} \neq \mathrm{sign}(\mathbf{g}_{t,ij})]$, which need not vanish. Consequently the finalized $\mathcal{O}(1/\sqrt{T})$ guarantee holds for the gradient-sign update $\mathbf{S}_t = \mathrm{sign}(\widetilde{\mathbf{G}}_t)$, whose direction never invokes the polar factor, and not for the LMO-based direction $\mathrm{sign}(\mathrm{polar}(\mathbf{M}_t))$ to which it is attributed.

On the instance of Theorem 1 the residual R_T is not merely unbounded: it exceeds $\mathcal{G}_T$. Consider the linear objective (8) with the 4×4 gradient $\mathbf{G}$ of Theorem 1. By Proposition 1 one has $\mathbf{g}_t \equiv \mathbf{G}$ and $\mathrm{polar}(\mathbf{M}_t) = \mathrm{polar}(\mathbf{G})$ at every iteration, so the oracle is deterministic and the inner-product identity $\mathbb{E}[\langle \mathbf{g}_t, \mathbf{D}_t\rangle \mid \mathbf{X}_t] = \frac{1}{\sqrt{mn}}\sum_{ij}|\mathbf{g}_{t,ij}|(1-2q_{ij,t})$ underlying (16) reduces to

$$\sum_{i,j}|\mathbf{G}_{ij}|\,(1-2q_{ij}) = \big\langle \mathbf{G}, \mathrm{sign}(\mathrm{polar}(\mathbf{G}))\big\rangle = -\tfrac{42468}{103} < 0. \tag{17}$$

Therefore $\sum_{ij}|\mathbf{G}_{ij}|\, q_{ij} > \frac{1}{2}\|\mathbf{G}\|_1$, and at every iteration

$$\frac{2}{\sqrt{mn}}\sum_{i,j}|\mathbf{G}_{ij}|\, q_{ij} \;>\; \frac{\|\mathbf{G}\|_1}{\sqrt{mn}} \;=\; \mathcal{G}_t. \tag{18}$$

The unestimated term R_T in (16) thus strictly exceeds the left-hand side, so the bound reduces to $\mathcal{G}_T \leq (\text{nonnegative}) + R_T$ with $R_T > \mathcal{G}_T$ and imposes no constraint on $\mathcal{G}_T$. In agreement, (17) yields the exact per-step increase $f(\mathbf{X}_t) - f(\mathbf{X}_{t-1}) = -\eta\langle \mathbf{G}, \mathbf{D}_t\rangle = \frac{\eta}{4}\cdot\frac{42468}{103} > 0$ for every $\eta > 0$, which is the divergence established in Theorem 1.

A.8 Extended comparison with S-Muon

Taking the norms dual to the Ky Fan k-norms, Kravatskiy et al. (2025) obtain the *Fanion* family, whose updates $\sum_{i\leq k}\mathbf{u}_i\mathbf{v}_i^\top$ interpolate between the rank-one step of the nuclear norm and Muon's full-rank $\mathbf{U}\mathbf{V}^\top$ at $k = \min(m,n)$; a conic combination of LMOs is again an LMO, for the norm dual to the corresponding combination of dual norms. Their S-Muon is one such combination, $\tau\,\mathbf{U}\mathbf{V}^\top + (1-\tau)\,c\,\mathrm{sign}(\mathbf{M}_t)$ with fixed $\tau \in [0,1]$, $c > 0$ (their notation differs; we reserve α for compressor contraction and η for the learning rate). It is the sharpest available contrast with this paper: the sign enters *inside* the oracle, so the step is still an LMO for an explicit norm and inherits the convergence theory of one, where our three placements act *around* the oracle. That the composed object is an LMO for no norm at all is not an assumption but a consequence of Theorems 1–3: an oracle for any norm returns $\mathbf{D}$ with $\langle \mathbf{G}, \mathbf{D}\rangle = \|\mathbf{G}\|_{\mathrm{dual}} \geq 0$, since $\mathbf{0}$ lies in every unit ball, and each of those theorems exhibits a $\mathbf{G}$ at which the composed step makes the inner product strictly negative. A caution from the same work applies to us directly: they exhibit an LMO method (rank-one Neon) markedly worse than Muon in practice despite sharing its convergence asymptotics in the bounds of Kovalev (2025) and Riabinin et al. (2025), from which our own guarantee descends. A rate of the form $\mathcal{O}(T^{-1/2})$ is a warrant for running a method, not a prediction of the performance it will attain.

A.9 Proof of Theorem 4 (Divergence of EF21-SignMuon)

Recall from the main text that EF21-SignMuon (Algorithm A3) does not descend along the LMO output $\mathbf{D}_t = \mathrm{polar}(\tilde{\mathbf{M}}_t)$ itself, but along the error-feedback estimate $\mathbf{d}_t^{\mathrm{est}}$ of (10), followed by $\mathbf{X}_t = \mathbf{X}_{t-1} - \eta\,\mathbf{d}_t^{\mathrm{est}}$. A *single* magnitude α_t rescales the signs of *all* entries at once, and it is this coupling that the counterexample exploits. We now prove Theorem 4.

Proof idea *The mechanism, in one dimension.* Suppose a scalar EF21 estimator must track a target that alternates between $+1$ and -1 while a *second* coordinate, sharing the same magnitude α_t, must track a small constant $-\varepsilon$. The alternating coordinate never permits the residual to shrink, so α_t remains at $\Theta(1)$; the small coordinate is then moved by $\pm\Theta(1)$ at every step and can only oscillate about its target, never settle on it. Which side of the target it spends more time on is decided by the *phase* of that oscillation, and a one-bit sign carries no information by which a phase could be corrected. If the phase is unfavourable, the time-average of the estimate has the sign *opposite* to $-\varepsilon$, and the iterate it drives runs off to infinity.

The construction below realizes exactly this with a 2×2 LMO target: the off-diagonal is large and flips sign every step, the diagonal is a small constant, and the shared α_t couples them. The result is a period-two limit cycle in the estimator whose $(2,2)$-average is positive while every target value is negative, so $(\mathbf{X}_t)_{22}$ increases without bound (Figure 1, right). The mechanism is not degeneracy: the momentum matrices driving the divergent tail have condition number $\frac{16}{9}$ throughout.

Why the phase has to be set. The unfavourable phase is not the one the method falls into by itself. Started from $\mathbf{d}_0^{\mathrm{est}} = \mathbf{0}$, the alternating targets alone drive the estimator to a *different* period-two cycle, one whose diagonal average has the *correct* sign (Lemma 2). The recursion (10) is piecewise affine, the pieces indexed by the sign pattern of the residual, and these two cycles reside in different pieces. The two-step preamble in the target sequence below serves a single purpose: it moves the estimator into the piece that contains the divergent cycle. That is why the counterexample requires a preamble at all, and it serves no other end.

Why the parameters drop out. The step size and the smoothness constant only rescale the orbit, so we may fix $\eta = 1$ and one value of L (Part 1). And the divergence depends on the estimator's response to a fixed sequence of *targets*; momentum affects only how the gradients that produce those targets are sourced. Inverting the momentum filter therefore settles every (μ, variant) at once.

Accordingly the proof has three parts, and they are independent. Part 1 rescales the problem to remove L and η. Part 2 is the dynamical core and is finite arithmetic: for one fixed sequence of LMO targets, the estimator enters a limit cycle whose diagonal has the wrong sign. Part 3 is an existence argument only: it exhibits a smooth function whose gradients generate that target sequence for every μ and either momentum variant. A reader willing to grant that some smooth objective produces the targets can stop after Part 2.

Proof

Part 1: rescaling.

Lemma 1 (Scale reduction) *If $\tilde{f}$ is $\tilde{L}$-smooth with EF21-SignMuon iterates $\tilde{\mathbf{X}}_t$ at step size 1, then $f(\mathbf{X}) := \frac{L\eta^2}{\tilde{L}}\,\tilde{f}(\mathbf{X}/\eta)$ is L-smooth and its run at step size η (same μ, same variant, from $\mathbf{0}$) satisfies $\mathbf{X}_t = \eta\,\tilde{\mathbf{X}}_t$ and $f(\mathbf{X}_t) = \frac{L\eta^2}{\tilde{L}}\,\tilde{f}(\tilde{\mathbf{X}}_t)$.*

Proof. $\nabla f(\mathbf{X}) = \frac{L\eta}{\tilde{L}}\nabla\tilde{f}(\mathbf{X}/\eta)$, so f is L-smooth. Suppose $\mathbf{X}_s = \eta\tilde{\mathbf{X}}_s$ for $s < t$: the two gradients then differ by the positive factor $\frac{L\eta}{\tilde{L}}$, which $\mathbf{M}_t$ and $\tilde{\mathbf{M}}_t$ (positive combinations of past gradients) inherit and polar discards. Hence $\mathbf{D}_t$ and $\mathbf{d}_t^{\text{est}}$ match the normalized run, and $\mathbf{X}_t = \eta\tilde{\mathbf{X}}_t$. ■ So it suffices to exhibit, for each (μ, variant), a C^∞ $\tilde{L}$-smooth $\tilde{f}$ on which the *normalized* run ($\eta = 1$) obeys (11); that $\tilde{f}$ may be taken bounded below (Assumption 1) is shown afterwards (Remark 1). We fix $\eta = 1$ from now on.

Part 2: the estimator's limit cycle. The dynamical core of the proof is the behavior of the estimator (10) on the fixed target sequence

$$\mathbf{D}_1 = \mathbf{S}_1, \quad \mathbf{D}_2 = \mathbf{S}_2, \quad \mathbf{D}_t = \bar{\mathbf{D}}^{(-1)^t}\ (t \geq 3), \tag{19}$$

$$\begin{gathered}\bar{\mathbf{D}}^{\pm} = \begin{pmatrix} 7/25 & \pm 24/25 \\ \pm 24/25 & -7/25 \end{pmatrix}, \\ \mathbf{S}_1 = \begin{pmatrix} -4/5 & 3/5 \\ -3/5 & -4/5 \end{pmatrix}, \quad \mathbf{S}_2 = \begin{pmatrix} 3/5 & -4/5 \\ 0 & 0 \end{pmatrix}.\end{gathered} \tag{20}$$

The divergence originates in the tail ($t \geq 3$): the reflections $\bar{\mathbf{D}}^{\pm}$ share the diagonal $(\frac{7}{25}, -\frac{7}{25})$, while their off-diagonal $\pm\frac{24}{25}$ reverses sign at every step. The rotation $\mathbf{S}_1$ and the rank-one $\mathbf{S}_2$ form a two-step preamble. Only $\mathbf{S}_2$ needs a word: at a rank-deficient argument the spectral-ball LMO is not unique, and $\mathbf{S}_2$ is the value returned by the rank-r convention of the main text, which is also what our implementation computes. The next lemma says what the preamble is for, and that something like it is unavoidable.

Lemma 2 (Two cycles, and only one diverges) *Write $\bar{\mathbf{D}}^{\pm} = \begin{pmatrix} a & \pm b \\ \pm b & -a \end{pmatrix}$ with $a^2 + b^2 = 1$ and $0 < a < b$. On the purely alternating targets $\mathbf{D}_t = \bar{\mathbf{D}}^{(-1)^t}$ ($t \geq 1$), the recursion (10) started from $\mathbf{d}_0^{\text{est}} = \mathbf{0}$ enters a period-two cycle immediately, namely $\mathbf{d}_t^{\text{est}} = \frac{a+b}{2}\begin{pmatrix} 1 & -1 \\ -1 & -1 \end{pmatrix}$ for odd t and $\frac{b-a}{2}\begin{pmatrix} -1 & 1 \\ 1 & 1 \end{pmatrix}$ for even t. Its $(2,2)$-average is $-\frac{a}{2}$: the* same *sign as every target value $-a$, at half the magnitude. That cycle is harmless.*

Proof. Put $m = \frac{a+b}{2}$. The residual $\bar{\mathbf{D}}^{-} - \mathbf{0}$ has signs $\begin{pmatrix} + & - \\ - & - \end{pmatrix}$ and mean modulus m, so $\mathbf{d}_1^{\text{est}} = m\begin{pmatrix} 1 & -1 \\ -1 & -1 \end{pmatrix}$. Next, $\bar{\mathbf{D}}^{+} - \mathbf{d}_1^{\text{est}} = \begin{pmatrix} a-m & b+m \\ b+m & m-a \end{pmatrix}$ has signs $\begin{pmatrix} - & + \\ + & + \end{pmatrix}$ (as $a < m$) and mean modulus b, giving $\mathbf{d}_2^{\text{est}} = \frac{b-a}{2}\begin{pmatrix} -1 & 1 \\ 1 & 1 \end{pmatrix}$. Repeating once returns $\mathbf{d}_1^{\text{est}}$. The average of the two $(2,2)$-entries is $\frac{1}{2}\left(-m + \frac{b-a}{2}\right) = -\frac{a}{2}$. ■

So the alternation by itself does not diverge, and no choice of (a, b) makes it do so. The divergent orbit is a *second* period-two cycle of the same recursion, distinguished by its residuals having a *uniform* sign pattern ($\begin{pmatrix} + & + \\ + & + \end{pmatrix}$ and $\begin{pmatrix} - & - \\ - & - \end{pmatrix}$) where the harmless cycle's are mixed. Since (10) is affine on each sign-pattern cell, the two cycles cannot be joined by the dynamics; the preamble exists solely to place $\mathbf{d}_2^{\text{est}}$ in the cell of the divergent one, which is what Lemma 3 verifies.

Lemma 3 (Wrong-sign limit cycle) *On the targets (19), the recursion (10) from $\mathbf{d}_0^{\text{est}} = \mathbf{0}$ enters at $t = 3$ the exact period-two cycle $\mathbf{d}_t^{\text{est}} = \mathbf{d}_B$ (odd t), $\mathbf{d}_A$ (even t), where*

$$\mathbf{d}_B = \frac{1}{200}\begin{pmatrix} -61 & -201 \\ -61 & -61 \end{pmatrix}, \qquad \mathbf{d}_A = \frac{1}{200}\begin{pmatrix} 131 & -9 \\ 131 & 131 \end{pmatrix}. \tag{21}$$

Its $(2,2)$-entry averages $\frac{1}{2}\big((\mathbf{d}_A)_{22} + (\mathbf{d}_B)_{22}\big) = +\frac{7}{40}$, opposite in sign to every target value $(\mathbf{D}_t)_{22} = -\frac{7}{25}$.

Proof. Substituting (19) into (10) gives the values

$$\begin{array}{c|cccc} t & 1 & 2 & 3 & 4 \text{ (then 2-periodic)} \\ \hline \alpha_t & \frac{7}{10} & \frac{21}{20} & \frac{131}{200} & \frac{24}{25} \\ \mathbf{d}_t^{\text{est}} & \frac{1}{10}\left(\begin{smallmatrix} -7 & 7 \\ -7 & -7 \end{smallmatrix}\right) & \frac{1}{20}\left(\begin{smallmatrix} 7 & -7 \\ 7 & 7 \end{smallmatrix}\right) & \mathbf{d}_B & \mathbf{d}_A \end{array} \tag{22}$$

Each residual $\mathbf{D}_t - \mathbf{d}_{t-1}^{\text{est}}$ has strictly nonzero entries, so the signs are unambiguous; e.g. at $t = 3$ it is $\frac{1}{100}\left(\begin{smallmatrix} -7 & -61 \\ -131 & -63 \end{smallmatrix}\right)$, all negative. From $t = 3$ the targets are 2-periodic and the pair $(\mathbf{d}_B, \mathbf{d}_A)$ reproduces itself: $\bar{\mathbf{D}}^+ - \mathbf{d}_B$ and $-(\bar{\mathbf{D}}^- - \mathbf{d}_A)$ are both entrywise positive with mean $\frac{24}{25}$, so $\mathbf{d}_B + \frac{24}{25}\mathbf{J} = \mathbf{d}_A$ and $\mathbf{d}_A - \frac{24}{25}\mathbf{J} = \mathbf{d}_B$ ($\mathbf{J}$ all-ones). ■

Since $\mathbf{X}_t = \mathbf{X}_{t-1} - \mathbf{d}_t^{\text{est}}$ (recall $\eta = 1$), over one period the $(2,2)$-coordinate changes by $-(\mathbf{d}_A + \mathbf{d}_B)_{22} = -\frac{7}{20}$. Consequently a term $-\gamma W_{22}$ in $\tilde{f}$, with $\gamma := \frac{7}{12}$, increases by $\gamma \cdot \frac{7}{20} = \frac{49}{240}$ per period. This is the divergence, provided a genuine smooth function produces the targets (19); Part 3 constructs one.

Part 3: realization by a smooth function. Fix (μ, variant) and set $\gamma = \frac{7}{12}$, $\nu = \frac{1}{1+2\mu}$, and

$$\begin{aligned} A &= \frac{1+\mu}{1-\mu} \text{ (standard)}, \\ A &= \frac{1+\mu}{(1-\mu)(1+2\mu)} \text{ (Nesterov)}. \end{aligned} \tag{23}$$

We build $\tilde{f} = g + \sum_{k=1}^{3} b_k$ from a periodic-plus-linear *field* g and three *localized corrections* b_k, all explicit.

The field is

$$g(\mathbf{W}) = -\gamma\, W_{22} + A\, \Phi_1(W_{12}) + A\, \Phi_2(W_{21}), \tag{24}$$

where $\Phi_i(w) = \int_0^w \psi_i$ and $\psi_i \colon \mathbb{R} \to [-1,1]$ is a fixed C^∞, p_i-periodic function with $\int_0^{p_i} \psi_i = 0$,

$$p_1 = \tfrac{21}{20}, \quad p_2 = \tfrac{7}{20}, \qquad \delta = \tfrac{1}{100},$$

equal to $+1$ on $[\rho_i^+ - \delta, \rho_i^+ + \delta]$ and -1 on $[\rho_i^- - \delta, \rho_i^- + \delta]$ (mod p_i), where

$$\rho_1^+ = \tfrac{131}{200},\ \rho_1^- = \tfrac{140}{200}; \qquad \rho_2^+ = \tfrac{61}{200},\ \rho_2^- = 0.$$

The two bands are disjoint mod p_i (their centers are $\frac{9}{200} > 2\delta$ apart), so such a ψ_i exists; the zero-mean condition, met by balancing the rest of the period, makes Φ_i periodic (hence bounded). On each band $\Phi_i' = \psi_i = \pm 1$ exactly.

The corrections pin the first three gradients. Fix once a C^∞ cutoff $\phi \colon \mathbb{R} \to [0,1]$ with $\phi \equiv 1$ on $[0, \frac{1}{2}]$ and $\phi \equiv 0$ on $[1, \infty)$, put $r = \frac{1}{50}$, and set

$$b_k(\mathbf{W}) = \big\langle \mathbf{C}_k,\, \mathbf{W} - \mathbf{Z}_k \big\rangle\, \phi\big(\|\mathbf{W} - \mathbf{Z}_k\|_F / r\big), \tag{25}$$

centered at the first three iterates (from Lemma 3)

$$\mathbf{Z}_1 = \mathbf{0}, \quad \mathbf{Z}_2 = \tfrac{1}{10}\left(\begin{smallmatrix} 7 & -7 \\ 7 & 7 \end{smallmatrix}\right), \quad \mathbf{Z}_3 = \tfrac{1}{20}\left(\begin{smallmatrix} 7 & -7 \\ 7 & 7 \end{smallmatrix}\right).$$

Each b_k is C^∞, supported in $\{\|\mathbf{W} - \mathbf{Z}_k\|_F \le r\}$; since its linear factor vanishes at $\mathbf{Z}_k$ while $\phi(0) = 1$, $\nabla b_k(\mathbf{Z}_k) = \mathbf{C}_k$. We choose $\mathbf{C}_k := \hat{\mathbf{G}}_k - \nabla g(\mathbf{Z}_k)$ with the explicit $\hat{\mathbf{G}}_k$ of (26) below, so that $\nabla \tilde{f}(\mathbf{Z}_k) = \hat{\mathbf{G}}_k$. Every term of $\tilde{f}$ has a bounded Hessian, so $\tilde{f}$ is C^∞ and $\tilde{L}$-smooth for a finite $\tilde{L}(\mu, \text{variant})$.

Lemma 4 (Realization) *For every $\mu \in [0,1)$ and either variant, EF21-SignMuon on this $\tilde{f}$ ($\eta = 1$, from $\mathbf{0}$) generates exactly the targets* (19).

Proof. *The required gradients.* Momentum is a causal, positive-weight linear filter of the gradients, and we invert it. Define the *transient gradients*

$$\hat{\mathbf{G}}_t = \frac{\mathbf{M}_t - \mu \mathbf{M}_{t-1}}{1-\mu} \quad (t = 1,2,3;\ \mathbf{M}_0 = \mathbf{0}) \tag{26}$$

and the *field gradient* $\mathbf{G}^\pm = \left(\begin{smallmatrix} 0 & \pm A \\ \pm A & -\gamma \end{smallmatrix}\right)$, where the prescribed buffer values $\mathbf{M}_{1,2,3}$ are given below. With $\bar{\mathbf{M}}^\pm = \left(\begin{smallmatrix} 0 & \pm 1 \\ \pm 1 & -\gamma \end{smallmatrix}\right)$, the factorization

$$\begin{gathered} \bar{\mathbf{M}}^\pm = \bar{\mathbf{D}}^\pm \left(\begin{smallmatrix} 24/25 & \mp 7/25 \\ \mp 7/25 & 337/300 \end{smallmatrix}\right) \\ \text{(second factor } \succ 0,\ \det = 1) \end{gathered} \tag{27}$$

shows $\operatorname{polar}(\bar{\mathbf{M}}^\pm) = \bar{\mathbf{D}}^\pm$, with singular values $\frac{3}{4}, \frac{4}{3}$ (the condition number $\frac{16}{9}$ of the idea).

Standard momentum ($\tilde{\mathbf{M}}_t = \mathbf{M}_t$). Take $\mathbf{M}_1 = \mathbf{S}_1, \mathbf{M}_2 = \mathbf{S}_2, \mathbf{M}_3 = \bar{\mathbf{M}}^-$; then $\hat{\mathbf{G}}_{1,2,3}$ are the explicit matrices (26). Feeding $\mathbf{G}^{(-1)^t}$ for $t \geq 4$ keeps $\mathbf{M}_t = \bar{\mathbf{M}}^{(-1)^t}$, because $(1-\mu)\mathbf{G}^{\pm} = \bar{\mathbf{M}}^{\pm} - \mu\bar{\mathbf{M}}^{\mp}$ with A as in (23). As $\operatorname{polar}(\mathbf{S}_1) = \mathbf{S}_1$ (orthogonal) and $\operatorname{polar}(\mathbf{S}_2) = \mathbf{S}_2$ (rank one), (27) yields the targets (19).

Nesterov momentum ($\tilde{\mathbf{M}}_t = (1+\mu)\mathbf{M}_t - \mu\mathbf{M}_{t-1}$). Here the choice of an *orthogonal* $\mathbf{S}_1$ earns its keep. Nesterov is a two-tap filter, so prescribing the tail fixes $\mathbf{M}_3$ and leaves only the preamble to absorb the mismatch at $t = 3$; and the set of matrices with a given polar factor $\mathbf{D}$ is $\{\mathbf{DH} : \mathbf{H} \succ 0\}$, which is three-dimensional when $\mathbf{D}$ is orthogonal but only one-dimensional (a positive scalar) when $\mathbf{D}$ has rank one. A preamble of two rank-one targets leaves too little freedom: we checked that it admits no realization once $\mu \gtrsim 0.3$. One orthogonal target suffices, and the single scalar T below is the freedom being spent. Take $\mathbf{M}_1 = \frac{1}{1+\mu}\mathbf{S}_1\mathbf{H}_1$, $\mathbf{M}_2 = \frac{1}{1+\mu}(\mathbf{S}_2 + \mu\mathbf{M}_1)$, $\mathbf{M}_3 = \bar{\mathbf{M}}'^-$ and $\mathbf{M}_t = \bar{\mathbf{M}}'^{(-1)^t}$ ($t \geq 4$), where $\bar{\mathbf{M}}'^{\pm} = \left(\begin{smallmatrix} 0 & \pm\nu \\ \pm\nu & -\gamma \end{smallmatrix}\right)$, $\mathbf{H}_1 = \operatorname{diag}(1, 1+T)$, and, for $\mu > 0$,

$$T = \left(\tfrac{140(1+\mu)^2}{1+2\mu} - 44\right)\tfrac{1+\mu}{117\mu} > 0.$$

(At $\mu = 0$ the Nesterov rule reads $\tilde{\mathbf{M}}_t = \mathbf{M}_t$ and is the standard case already treated, so nothing is left to prove there.) Then $\tilde{\mathbf{M}}_1 = \mathbf{S}_1\mathbf{H}_1$ and $\tilde{\mathbf{M}}_2 = \mathbf{S}_2$ have polar factors $\mathbf{S}_1, \mathbf{S}_2$, and $\tilde{\mathbf{M}}_t = \bar{\mathbf{M}}^{(-1)^t}$ for $t \geq 4$ (since $\nu(1+2\mu) = 1$). The one nontrivial step is $t = 3$: with $\mathbf{H}_3 := \bar{\mathbf{D}}^-\tilde{\mathbf{M}}_3$ (so $\tilde{\mathbf{M}}_3 = \bar{\mathbf{D}}^-\mathbf{H}_3$, the reflection being an involution), the stated T is exactly the value making $\mathbf{H}_3$ symmetric, and $\mathbf{H}_3 \succ 0$ for all $\mu \in (0,1)$: under $\mu = \frac{s}{1+s}$ ($s > 0$) the numerators of its two leading minors are polynomials in s with nonnegative coefficients. Hence $\operatorname{polar}(\tilde{\mathbf{M}}_3) = \bar{\mathbf{D}}^-$, completing (19).

The function delivers these gradients. It remains to verify $\nabla\tilde{f}(\tilde{\mathbf{X}}_{t-1}) = \hat{\mathbf{G}}_t$ for $t \leq 3$ and $= \mathbf{G}^{(-1)^t}$ for $t \geq 4$. By Lemma 3 the iterates $\tilde{\mathbf{X}}_0, \tilde{\mathbf{X}}_1, \tilde{\mathbf{X}}_2$ are exactly the centers $\mathbf{Z}_1, \mathbf{Z}_2, \mathbf{Z}_3$, where $\nabla\tilde{f} = \hat{\mathbf{G}}_{1,2,3}$ by construction. The three balls are disjoint ($\|\mathbf{Z}_j - \mathbf{Z}_k\|_F \geq \frac{7}{10} > 2r$), and every later iterate has $(1,2)$-entry $\geq \frac{131}{200}$ while the centers have it ≤ 0, so no ball is re-entered. For $t \geq 4$ the query lies in the field, where

$$\nabla g(\mathbf{W}) = \begin{pmatrix} 0 & A\psi_1(W_{12}) \\ A\psi_2(W_{21}) & -\gamma \end{pmatrix}.$$

The period shift $\tilde{\mathbf{X}}_{t+2} - \tilde{\mathbf{X}}_t = -(\mathbf{d}_A + \mathbf{d}_B)$ advances W_{12} by $+p_1$ and W_{21} by $-p_2$ (one full period each), so the off-diagonal coordinates keep the residues ρ_i^+ at odd indices and ρ_i^- at even indices; there $\psi_i = \pm 1$, giving $\nabla g = \mathbf{G}^{(-1)^t}$. ■

Proof of Theorem 4. By Lemma 4 the normalized run produces the targets (19), so by Lemma 3 its estimator locks onto the wrong-sign cycle and $\tilde{\mathbf{X}}_{t+2} - \tilde{\mathbf{X}}_t$ is a constant shift. Along it Φ_1, Φ_2 return to their values and the corrections b_k vanish, so only the linear term acts: $\tilde{f}(\tilde{\mathbf{X}}_{t+2}) - \tilde{f}(\tilde{\mathbf{X}}_t) = -\gamma(-\frac{7}{20}) = \frac{49}{240} > 0$. By Lemma 1, f then obeys (11) with $c = \frac{49}{240\tilde{L}}$. As $(L, \eta, \mu, \text{variant})$ were arbitrary, no rule $\eta = \eta(L, \mu)$ can prevent divergence. ■

Remark 1 (Boundedness below) *Only the linear term $-\gamma W_{22}$ makes $\tilde{f}$ unbounded below, and only as $W_{22} \to +\infty$, a region the iterates never enter, since $(\tilde{\mathbf{X}}_t)_{22} \leq \frac{7}{10}$ throughout (it decreases after the transient). Replacing $-\gamma W_{22}$ by any C^∞ function that agrees with it on $\{W_{22} \leq 1\}$ and is constant on $\{W_{22} \geq 2\}$ therefore leaves the whole trajectory, and (11) with it, unchanged while rendering $\tilde{f}$ bounded below (Assumption 1); the theorem is stated with this modification in force.*

Remark 2 (Verification) *The construction is checked in two independent ways: symbolically, in exact rational arithmetic, and numerically, by running the float64 reference implementation of Algorithm A3 on the assembled $\tilde{f}$. The right panel of Figure 1 confirms that at $\mu = 0$ EF21-SignMuon is the only one of the eight methods that diverges, the others (SignMuon, MuonUSign, MuonSign, EF21-MuonUSign, EF21-MuonSign, SignSGD, Muon) staying bounded; Figure A1 confirms that EF21-SignMuon diverges at the exact rate $\frac{49}{480}$ for every $\mu \in \{0, \frac{1}{2}, 0.9, 0.95, 0.99\}$ under both standard and Nesterov momentum, as the reduction predicts.*

A.10 Convergence of EF21-MuonUSign and EF21-MuonSign

We do not analyse the error-feedback methods from scratch. EF21-MuonUSign and EF21-MuonSign are *exact instances* of EF21-Muon (Gruntkowska et al. 2025, Algorithm 3), already analysed in the layer-wise, stochastic, federated setting. Three conditions must be verified before its guarantees transfer: our step is their LMO step, our loop is their loop (Proposition 2), and our messages come from contractive compressors (Lemma 5). Only the last is non-trivial. Table A1 is the change of variables.

Notation and constants. For the layer tuple $\mathbf{X} = [\mathbf{X}_1, \ldots, \mathbf{X}_p]$, $\mathbf{X}_i \in \mathbb{R}^{m_i \times n_i}$, of the Problem Statement write $d_i := m_i n_i$, $r_i := \min(m_i, n_i)$, $d_{\max} := \max_i d_i$; a second subscript selects a layer ($\mathbf{X}_{t,i}$, $\mathbf{g}_{t,i}$). We use $\|\mathbf{Y}\|_{2\to 2} \leq \|\mathbf{Y}\|_F \leq \|\mathbf{Y}\|_* \leq \sqrt{\operatorname{rank}\mathbf{Y}}\,\|\mathbf{Y}\|_F$. Smoothness constants: L_i for f and $L_{i,j}$ for f_j in Assumption 2, with $\tilde{L}_i^2 := \frac{1}{N}\sum_j L_{i,j}^2$ and $L := \max_i L_i$; in the (L^0, L^1) form the pairs are again per layer for f and per layer and client for f_j, with $L^1_{i,\max} := \max_j L^1_{i,j}$.

Assumption 1 (Stochastic gradient) *Each client's stochastic gradient is unbiased, $\mathbb{E}_\xi[\nabla f_j(\mathbf{X}; \xi)] = \nabla f_j(\mathbf{X})$, with bounded variance $\mathbb{E}_\xi\|\nabla f_j(\mathbf{X}; \xi) - \nabla f_j(\mathbf{X})\|_*^2 \leq \sigma^2$.*

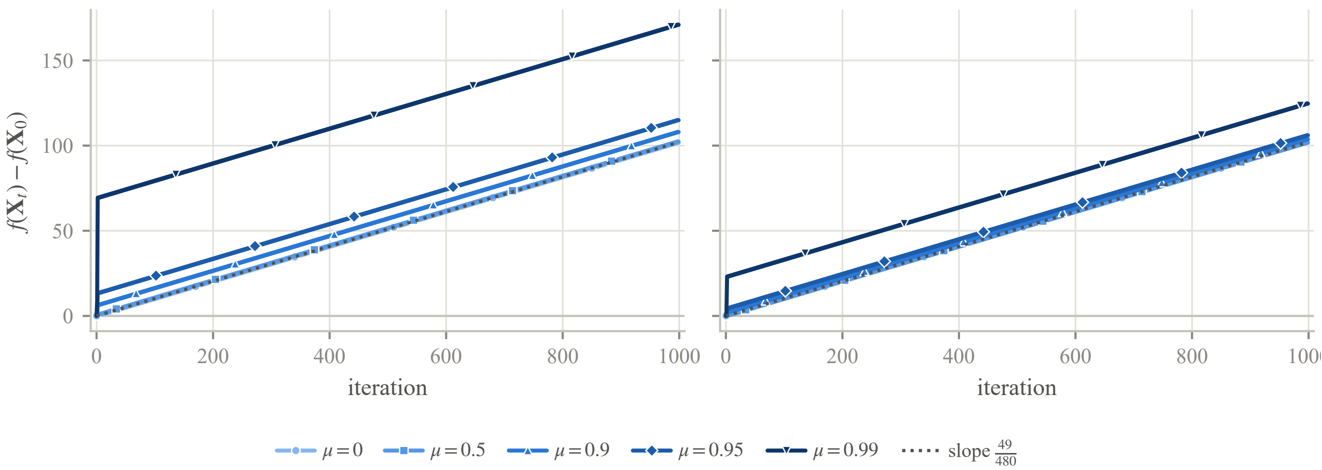

Figure A1: Momentum does not prevent the divergence of EF21-SignMuon. For each momentum coefficient μ and variant, EF21-SignMuon is run on the corresponding instance of Theorem 4 and $f(\mathbf{X}_t) - f(\mathbf{X}_0)$ is plotted; *left:* standard momentum, *right:* Nesterov. Every setting diverges at the common rate $\frac{49}{480}$ (dotted). Subtracting $f(\mathbf{X}_0)$ removes the only genuinely μ-dependent offset (the field constant $A(\mu)$ scales a bounded periodic term); what remains is a bounded transient, largest as $\mu \to 1$, on top of the shared linear divergence.

Assumptions 1–1 are Assumptions 1–2 and 6–10 of Gruntkowska et al. (2025) with the layer norms taken spectral; our variance bound is stated in the nuclear norm and implies theirs via $\|\cdot\|_F \leq \|\cdot\|_*$. The clause $f_j \geq f_j^*$ of Assumption 1 is needed only for Corollary 2.

Main result

Theorem 5 (Convergence of the EF21 methods) *Run the federated Algorithm A9 with the EF21 uplink and EMA momentum* $\mu \in [0, 1)$. *Then:*

(i) (smooth; both methods) *under Assumptions 1–1, with the sharp learning rate* $\eta_{t,i} = \gamma_i \|\mathbf{g}_{t,i}\|_*$ *and tuned* (γ_i, μ)*, both reach* $\frac{1}{T}\sum_{t<T} \mathbb{E}\|\nabla f(\mathbf{X}_t)\|_*^2 = \mathcal{O}(T^{-1/2})$ *(Corollary 1);*

(ii) (generalized smooth; uplink only) *under* (L^0, L^1)*-smoothness, EF21-MuonUSign with a plain* constant *learning rate reaches* $\min_{t \leq T} \sum_i \mathbb{E}\|\nabla_i f(\mathbf{X}_t)\|_* = \mathcal{O}(T^{-1/4})$ *(Corollary 2).*

Both statements are written for a per-layer constant common to all layers ($\gamma_i \equiv \gamma$, $\eta_i \equiv \eta$); for unequal constants the left-hand sides carry the step-size weights of Corollaries 1–2. The majority-vote placements (Theorems 1–3) and EF21-SignMuon (Theorem 4) admit no such guarantee.

Part (i) is the rate announced in the main text (via $\min_t \leq$ average); the centralized Algorithms A6–A7, and hence $\mathbf{X}_t$ in (i), are the case $N = 1$.

The reduction

The step is their LMO step. For $\mathbf{G} = \mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^\top$ the framework's oracle over the spectral ball of radius τ is $\mathrm{lmo}(\mathbf{G}) = \mathbf{X} - \tau\mathbf{U}\mathbf{V}^\top$, which is our server step with $\tau = \eta_{t,i}$, since $\mathbf{D}_{t,i} = -A(\mathbf{g}_{t,i}) = \mathbf{U}_{t,i}\mathbf{V}_{t,i}^\top$:

$$\mathbf{X}_{t,i} = \mathbf{X}_{t-1,i} - \eta_{t,i}\,\mathbf{D}_{t,i} = \mathrm{lmo}_{\mathcal{B}(\mathbf{X}_{t-1,i}, \eta_{t,i})}(\mathbf{g}_{t,i}). \tag{28}$$

Their "sharp" step $\mathbf{X} - \gamma\,\mathbf{G}^\sharp$ uses $\mathbf{G}^\sharp = \|\mathbf{G}\|_*\mathbf{U}\mathbf{V}^\top$, so a constant γ_i *is* the schedule $\eta_{t,i} = \gamma_i\|\mathbf{g}_{t,i}\|_*$ and a constant $\eta_{t,i}$ is the plain Muon rate: both are learning-rate choices, not algorithmic modifications. The spectral norm-equivalence constants are $\underline{\rho}_i = 1$, $\bar{\rho}_i = \sqrt{r_i}$, defined by $\underline{\rho}_i\|\mathbf{Y}\|_{2\to 2} \leq \|\mathbf{Y}\|_F \leq \bar{\rho}_i\|\mathbf{Y}\|_{2\to 2}$. At rank-deficient $\mathbf{G}$ the LMO is non-unique, but any selection serves: writing $\boldsymbol{\Delta} := \mathrm{lmo}_{\mathcal{B}(\mathbf{X},\tau)}(\mathbf{G}) - \mathbf{X}$ for the displacement it produces, the analysis uses only $\langle \mathbf{G}, \boldsymbol{\Delta}\rangle = -\tau\|\mathbf{G}\|_*$ and $\|\boldsymbol{\Delta}\|_{2\to 2} \leq \tau$, with $\mathbf{G} = \mathbf{0}$ read as the zero step ($\mathbf{0}^\sharp = \mathbf{0}$).

The momentum is their momentum. Algorithm A9 uses $\mathbf{M}_t = \mu\mathbf{M}_{t-1} + (1-\mu)\mathbf{G}_t$, which is (5) and is the framework's momentum with $\beta = 1 - \mu$. Rescaling the momentum stream by a constant, as the heavy-ball convention does by the factor $1/(1-\mu)$, alters nothing: the EF21 recursion is positively homogeneous and the Muon LMO scale-invariant, so the factor leaves the iterates unchanged under a constant $\eta_{t,i}$ and is absorbed into γ_i under the sharp schedule. (The Nesterov branch is a different filter; see the end of this appendix.)

	This paper	EF21-Muon
clients / rounds	N, T	$n, K = T + 1$
iterate	$\mathbf{X}_t$	X^{t+1}
momentum	μ	$1 - \beta$
learning rate	$\eta_{t,i}$	radius t_i^t
norm equivalence	$1, \sqrt{r_i}$	$\underline{\rho}_i, \bar{\rho}_i$
uplink compr.	scaled sign	$\mathbb{B}_2(\alpha_D)$
downlink compr.	exact / sign	$\mathcal{I}$ / $\mathbb{B}_2(\alpha_P)^\dagger$
compression α	$1/d_{\max}$	α_D

Table A1: Change of variables from our notation to that of Gruntkowska et al. (2025). Layer norms $\|\cdot\|_{(i)}, \|\cdot\|_{(i)\star}$ are read as $\|\cdot\|_{2\to 2}, \|\cdot\|_*$ and smoothness constants $L_i^0, \tilde{L}_i^0$ as $L_i, \tilde{L}_i$; the index shift is Proposition 2. †Their Theorem 19 as printed asks for a server compressor in $\mathbb{B}(\alpha_P)$, contractive in the *layer* norm; the Euclidean class $\mathbb{B}_2(\alpha_P)$ that the scaled sign belongs to is licensed only by their Remark 23, at the $\bar{\rho}_i^2$ price charged in Corollary 1 and Remark 4.

Proposition 2 (Exact instance) *Fix μ, a learning-rate schedule, and the LMO selection above. Started from $\mathbf{M}_0^{(j)} = \mathbf{g}_0^{(j)} = \mathbf{g}_0 = \mathbf{0}$, $\mathbf{W}_0 = \mathbf{X}_0$, Algorithm A9 with the EF21 uplink and either downlink mode ($\mathcal{C}^\downarrow \in \{$exact, EF21-P$\}$) produces the same trajectory as Algorithm 3 of Gruntkowska et al. (2025) with spectral norms, scaled-sign worker compressors, identity/scaled-sign server compressor, $\beta = 1 - \mu$, and radii $t_i^k = \eta_{k,i}$, up to a one-step index shift $X^{t+1} = \mathbf{X}_t$.*

Proof The loops differ only in where the round is cut: they order it *step* $\to$ *downlink* $\to$ *gradient* $\to$ *uplink*, we order it *downlink* $\to$ *gradient* $\to$ *uplink* $\to$ *step*. Their iteration $k = 0$ is vacuous under our initialization: $\mathbf{g}_0 = \mathbf{0}$ gives $X^1 = X^0 = \mathbf{X}_0$ and a zero downlink residual, so $W^1 = \mathbf{X}_0$. Thereafter their iteration $k = t$ performs our round t verbatim, with the same momentum, the same compressed residual and the same LMO step (28), giving $X^{t+1} = \mathbf{X}_t$, $W^{t+1} = \mathbf{W}_t$ by induction. Running their method for $K = T + 1$ iterations therefore yields $\{\mathbf{X}_0, \mathbf{X}_0, \mathbf{X}_1, \ldots, \mathbf{X}_{T-1}\}$, and any average or minimum over these equals ours up to one duplicated nonnegative term. ■

Zero initialization also makes their initial-error constant Ψ^0 explicit: the $\|\mathbf{M}_0 - \mathbf{g}_0\|$ term vanishes and the gradient-deviation terms become $\|\nabla_i f(\mathbf{X}_0)\|$. The β^{-1} surviving in Ψ^0 enters only through the Ψ^0/T term, which $T^{-1/2}$ dominates.

The one real check: the scaled sign is a compressor The framework requires every transmitted message to originate in a *contractive compressor*: a map with $\mathbb{E}\|\mathcal{C}(\mathbf{Y}) - \mathbf{Y}\|^2 \le (1-\alpha)\|\mathbf{Y}\|^2$ for some $\alpha \in (0, 1]$ (Gruntkowska et al. 2025, Def. 1). This is precisely the property that fails for a bare sign and holds once it is scaled.

Lemma 5 (Contractivity of the scaled sign) *For every $\mathbf{Y} \in \mathbb{R}^{m\times n}$ ($d = mn$), the scaled sign $\mathcal{C}(\mathbf{Y}) = \mathrm{mean}(|\mathbf{Y}|)\,\mathrm{sign}(\mathbf{Y})$, with exact zeros resolved to ± 1 as in Section 4, satisfies*

$$\|\mathcal{C}(\mathbf{Y}) - \mathbf{Y}\|_F^2 = \|\mathbf{Y}\|_F^2 - \tfrac{1}{d}\|\mathbf{Y}\|_1^2 \le \left(1 - \tfrac{1}{d}\right)\|\mathbf{Y}\|_F^2, \tag{29}$$

so $\mathcal{C}$ is Euclidean-contractive with $\alpha = 1/d$. The identity holds for every draw of the random signs, not merely in expectation, and gives the exact contraction $\alpha(\mathbf{Y}) = \|\mathbf{Y}\|_1^2/(d\|\mathbf{Y}\|_F^2)$, which is $\Theta(1)$ for dense $\mathbf{Y}$ (e.g. $\to 2/\pi$ for i.i.d. Gaussian entries) and equals $1/d$ exactly at any 1-sparse $\mathbf{Y}$.

Proof Write $c := \|\mathbf{Y}\|_1/d$ and $s_{kl} = \pm 1$ for the transmitted signs. A nonzero entry contributes $(|Y_{kl}| - c)^2$ and a zero entry contributes $(c\,s_{kl})^2 = c^2$ whichever sign was drawn, so $\|\mathcal{C}(\mathbf{Y}) - \mathbf{Y}\|_F^2 = \|\mathbf{Y}\|_F^2 - 2c\|\mathbf{Y}\|_1 + c^2\|\mathbf{Y}\|_0 + c^2(d - \|\mathbf{Y}\|_0) = \|\mathbf{Y}\|_F^2 - 2c\|\mathbf{Y}\|_1 + dc^2$, which is (29) on substituting c. The bound then follows from $\|\mathbf{Y}\|_1 \ge \|\mathbf{Y}\|_F$, with equality exactly at the 1-sparse $\mathbf{Y}$; the Gaussian limit uses $\mathbb{E}|Y_{kl}| = \sqrt{2/\pi}\,\sigma$. ■

The $\|\mathbf{Y}\|_0$ terms cancel only under the randomized sign; otherwise the identity is sparsity-dependent and $1/d$ becomes an unattained infimum.

The scaling is what sustains the lemma. A bare sign contracts for no α at all: as $\|\mathbf{Y}\|_F \to 0$ on a fixed support, $\|\mathrm{sign}(\mathbf{Y}) - \mathbf{Y}\|_F \to \sqrt{\|\mathbf{Y}\|_0}$. That is the dividing line between our divergent and convergent methods: the majority-vote methods transmit unscaled signs of full quantities, the EF21 variants the scaled sign of a *residual*, at the cost of one extra scalar per layer per round. Per layer the uplink scaled sign lies in $\mathbb{B}_2(1/d_i) \subseteq \mathbb{B}_2(\alpha)$ with $\alpha := 1/d_{\max}$; the downlink is exact for EF21-MuonUSign ($\alpha_P = 1$) and the same scaled sign for EF21-MuonSign.

Transferred guarantees All requirements hold, so the framework's theorems apply through Table A1. We state the rates and stepsize rules; the explicit non-asymptotic bounds are Gruntkowska et al. (2025), Theorem 19 (smooth) and Theorem 24 ((L^0, L^1)), at these constants.

Corollary 1 (Smooth case; both methods) *Let Assumptions 1–1 hold. Run Algorithm A9 with the EF21 uplink, EMA momentum, and $\eta_{t,i} = \gamma_i\|\mathbf{g}_{t,i}\|_*$ with any γ_i below the per-layer threshold of Gruntkowska et al. (2025, Thm. 19) under Table A1. Then,* *applying the momentum tuning of Gruntkowska et al. (2025, Cor. 2) per layer (their Corollaries 1–2 state it for the layer-wise initialization and for $p = 1$ respectively; the combination is routine),*

$$\frac{1}{T}\sum_{t<T}\sum_{i=1}^{p} w_i\,\mathbb{E}\|\nabla_i f(\mathbf{X}_t)\|_*^2 = \mathcal{O}(T^{-1/2}),$$

with $w_i := \gamma_i/(\frac{1}{p}\sum_l \gamma_l)$; for a common γ_i all $w_i = 1$ and the left side is $\frac{1}{T}\sum_{t<T}\mathbb{E}\|\nabla f(\mathbf{X}_t)\|_*^2$.

Proof By Proposition 2 the run is an instance of Algorithm 3 with $K = T + 1$, and by Lemma 5 its compressors satisfy $\alpha_D = \alpha$ and $\alpha_P \in \{1, \alpha\}$ (EF21-MuonSign's Euclidean downlink is admissible by their Remark 23, which multiplies the α_P-terms of the threshold constant ζ_i by $\bar{\rho}_i^2 = r_i$). Substituting $\underline{\rho}_i = 1$, $\bar{\rho}_i = \sqrt{r_i}$ into Gruntkowska et al. (2025, Thm. 19) gives the threshold and the rate; the duplicated $\mathbf{X}_0$-term costs a factor $\frac{T+1}{T}$. ■

Since $\gamma_i \lesssim \zeta_i^{-1/2}$, the threshold shrinks polynomially in the layer dimension, through $\bar{\rho}_i = \sqrt{r_i}$ and the uplink $\alpha = 1/d_{\max}$, and for EF21-MuonSign by a further $\sqrt{r_i}$ on account of the compressed downlink. That additional factor is structural: the scaled sign is spectrally contractive for no parameter at all (Remark 4).

Corollary 2 **((L^0, L^1) case; EF21-MuonUSign)** *Let Assumption 2 hold in its (L^0, L^1) form. Run EF21-MuonUSign (exact downlink) with $S := T + 2$, momentum $\mu = 1 - S^{-1/2}$, and the* constant *per-layer rate $\eta_{t,i} \equiv \eta_i/S^{3/4}$ for any $\eta_i \le 1$ with $\eta_i^2\sqrt{r_i}\,(L^1_{i,\max})^2 = \mathcal{O}(1)$.*[1] *Then*

$$\min_{0\le t\le T}\sum_{i=1}^{p} v_i\,\mathbb{E}\|\nabla_i f(\mathbf{X}_t)\|_* = \mathcal{O}(T^{-1/4}), \quad v_i := \frac{\eta_i}{\frac{1}{p}\sum_l \eta_l}.$$

Proof Their Theorem 24 requires the identity server compressor, i.e. the exact downlink of EF21-MuonUSign (so EF21-MuonSign is excluded). Applying it with $K = T + 1$, $\beta = S^{-1/2}$ and Table A1 gives the schedule and the rate. ■

Under plain smoothness ($L^1 = 0$) every constraint above collapses to $\eta_i \le 1$. The *constant*-learning-rate EF21-MuonUSign we actually run therefore already converges at $\mathcal{O}(T^{-1/4})$; the sharp schedule yields only the faster $\mathcal{O}(T^{-1/2})$ in squared norm.

Scope The reduction covers the two error-feedback methods and no others, which matches our negative results. The majority-vote methods send unscaled signs and fall outside the framework. EF21-SignMuon compresses the LMO *output* rather than the gradient, so it tracks the non-Lipschitz polar factor and the momentum-tracking step breaks (Theorem 4). EF21-MuonSign gets the smooth guarantee but not the (L^0, L^1) one, whose theory assumes an uncompressed downlink.

The remaining gap is the Nesterov branch, which our language-model runs use (Appendix A.16): with $\beta := 1 - \mu$ it steers by $\tilde{\mathbf{M}}_t = (1-\beta)\mathbf{M}_t + \beta\mathbf{G}_t$ rather than by the buffer $\mathbf{M}_t$, a two-tap filter of the gradients where the framework presupposes a first-order recursion. The discrepancy is small and explicit: writing $\mathbf{n}_t := \mathbf{G}_{t,i} - \nabla_i f(\mathbf{X}_t)$, we have $\nabla_i f - \tilde{\mathbf{M}}_t = (1-\beta)(\nabla_i f - \mathbf{M}_t) - \beta\,\mathbf{n}_t$, and since $\mathbb{E}\langle\nabla_i f - \mathbf{M}_t, \mathbf{n}_t\rangle = -\beta\,\mathbb{E}\|\mathbf{n}_t\|_2^2$,

$$\mathbb{E}\|\nabla_i f - \tilde{\mathbf{M}}_t\|_2^2 \le (1-\beta)^2\,\mathbb{E}\|\nabla_i f - \mathbf{M}_t\|_2^2 + 3\beta^2\sigma_i^2.$$

The first term is exactly the deviation Gruntkowska et al. (2025, Thm. 19) already tracks, contracted rather than enlarged; the second is dominated by the $\beta\sigma_i^2$ term already in its bound. Nesterov should therefore degrade the constants rather than the rate. We state this as an expectation, not as a corollary, since the tracking recursion for $\|\tilde{\mathbf{M}}_{t+1} - \tilde{\mathbf{M}}_t\|$ would have to be redone as well.

Remark 3 (The cost of one bit) *The uplink $\alpha = 1/d_{\max}$ is what makes the threshold of Corollary 1 shrink with the dimension, this being the worst-case cost of one-bit messages, as it is for Top-1 compressors in Euclidean EF21 (Richtárik, Sokolov, and Fatkhullin 2021).* On the uplink *it is worst-case only: it requires a residual all of whose coordinates but one are negligible, whereas the momentum residual is dense with $\alpha(\mathbf{\Delta}) = \Theta(1)$ (Lemma 5), and the framework is stated to extend straightforwardly to iteration-dependent α (Gruntkowska et al. 2025, Rem. 12).*

The downlink of EF21-MuonSign does not inherit this. *Its residual is not a gradient difference refreshed by fresh stochasticity, but the output of the compressor's own recursion*

$$\mathbf{\Delta}_{t+1}^{\downarrow} = \mathbf{\Delta}_t^{\downarrow} - \eta_{t+1}\mathbf{D}_{t+1} - \operatorname{mean}|\mathbf{\Delta}_t^{\downarrow}|\operatorname{sign}(\mathbf{\Delta}_t^{\downarrow}),$$

in which every coordinate is corrected by the same *amount. Coordinates driven harder than that mean are never overtaken while the bulk holds the mean down, so the residual concentrates, and concentration is what drives α down. We observe this: $\alpha(\mathbf{\Delta}^{\downarrow})$ falls to 1.2×10^{-4} on the layers built from a zero initialization, nearly four orders of magnitude below its own uplink on the same layers (Section 5.3). That remains far above the $1/d = 4.2 \times 10^{-7}$ floor, so the dimension dependence of Corollary 1 is not attained; what is forfeited is the $\Theta(1)$ contraction that renders it harmless upstream.*

[1] These are the binding conditions of Gruntkowska et al. (2025, Thm. 24); the two remaining ones relax as T grows. Their printed second condition is tighter than the display its own proof establishes by a factor $K + 1$; we quote the latter.

Remark 4 (Why the sharp route is closed) *Corollary 1 reaches EF21-MuonSign only through the $\bar{\rho}_i^2$-penalized branch of Gruntkowska et al. (2025, Rem. 23), and this is forced. The sharp branch asks for a server compressor in $\mathbb{B}(\alpha_P)$, contractive in the* layer *norm, and the scaled sign is in no such class for any $\alpha_P > 0$. Take $\mathbf{Y}_\delta = (1-\delta)\mathbf{I}_n + \delta\mathbf{J}_n$, with $\mathbf{J}_n$ all ones, $0 < \delta < 1$ and $n \geq 2$. Every entry is positive, so $\operatorname{sign}(\mathbf{Y}_\delta) = \mathbf{J}_n$ and $\operatorname{mean}|\mathbf{Y}_\delta| = (1 + (n-1)\delta)/n$, whence*

$$\mathcal{C}(\mathbf{Y}_\delta) - \mathbf{Y}_\delta = (1-\delta)\big(\tfrac{1}{n}\mathbf{J}_n - \mathbf{I}_n\big),$$

with eigenvalues 0 and $-(1-\delta)$. So $\|\mathcal{C}(\mathbf{Y}_\delta) - \mathbf{Y}_\delta\|_{2\to 2} = 1 - \delta$ against $\|\mathbf{Y}_\delta\|_{2\to 2} = 1 + (n-1)\delta$, a ratio tending to 1 as $\delta \downarrow 0$: no $\alpha_P > 0$ survives, on any layer of rank above one. The framework's sufficient condition is unattainable for the same reason: by the norm-equivalence argument of Gruntkowska et al. (2025, App. D) a compressor in $\mathbb{B}_2(\alpha)$ is certified to lie in $\mathbb{B}(\alpha_P)$ only if $\alpha > 1 - 1/r_i$, i.e. $\alpha > 0.9987$ for our 768×3072 matrices, against an isotropic $2/\pi$. The factor $\bar{\rho}_i^2 = r_i$ in ζ_i is therefore not removable by a sharper analysis of this compressor, but only by replacing it, with random dropout ($\alpha_P = p$) or a Top-K SVD compressor ($\alpha_P = 1 - \sigma_{K+1}^2/\sigma_1^2$), both layer-norm contractive at a comparable budget (Gruntkowska et al. 2025, App. D), and neither strictly one-bit.

Remark 5 (From Muon to Gluon) *Only $(\underline{\rho}_i, \bar{\rho}_i) = (1, \sqrt{r_i})$ is spectral-norm-specific: Lemma 5 is Euclidean and the framework's theorems hold for arbitrary layer norms. A new geometry has to supply only its LMO and its norm-equivalence pair; both corollaries then hold with those constants in place of $(1, \sqrt{r_i})$, giving* EF21-GluonUSign *and* EF21-GluonSign *for the Gluon setting (Riabinin et al. 2025). Admissible geometries abound: $\ell_1 \to \ell_\infty$ for embeddings (Pethick et al. 2025a), the Schatten-p norms (Cesista 2025), the Ky Fan duals of the Fanion family (Kravatskiy et al. 2025), and, by the closure property of the last work, conic combinations of them. Such a combination adds the oracle outputs, $\sum_j \alpha_j \operatorname{lmo}_{(j)}$, so its unit ball is the Minkowski sum $\sum_j \alpha_j \mathcal{B}_{(j)}$ and its dual norm the corresponding sum $\sum_j \alpha_j \|\cdot\|_{(j)\star}$; since every element of that ball decomposes as $\sum_j \alpha_j \mathbf{Y}_j$ with $\|\mathbf{Y}_j\|_{(j)} \leq 1$, the mixture inherits $\bar{\rho}_i \leq \sum_j \alpha_j \bar{\rho}_i^{(j)}$. Convergence is thus a property of error feedback together with scaled-sign compression, not of the spectral geometry. The two ways to bring a sign into a Muon step stay cleanly apart: inside the oracle it changes the norm and the guarantee follows the norm; outside it, it is a compressor and the guarantee has to come from error feedback.*

We assume the exact spectral LMO, as do all analyses of Muon-type methods (Li and Hong 2025; Kovalev 2025; Riabinin et al. 2025; Gruntkowska et al. 2025); in practice it is the five-step iteration of Algorithm A1 (Amsel et al. 2025; Grishina, Smirnov, and Rakhuba 2025), with vector parameters and the last layer trained by AdamW as usual (Jordan et al. 2024). That substitution is not covered by the reduction either, but it is the mildest of the gaps: Shulgin et al. (2026) analyse the inexact Muon update directly and find the method tolerant of oracle error, which is consistent with our runs: the five-step approximation reproduces record #40 to within seed noise (Table 3).

A.11 Reproducibility Details

Why these four problems. Each benchmark answers a question the others cannot. The convex quadratic of Appendix A.12 is the only one whose gradient, smoothness constant and minimizer are known in closed form, so it is where the alignment the counterexamples attack can be measured rather than inferred. CIFAR-10 with a ResNet-18 (Krizhevsky and Hinton 2009) carries no such special property: we use it because it is the benchmark on which the sign-compression line of work is quoted (Bernstein et al. 2018, 2019; Karimireddy et al. 2019) and on which SignMuon's own concurrent proposal is evaluated (Mishra, Trivedi, and Kumar 2026), so it is where our methods can be placed against numbers other people have published; it is small enough to run every method at several step sizes and several seeds, which is what the claims about seed spread require. The federated split of the same data at $N = 11$ isolates the effect of the compressor from the effect of the architecture, since only the communication pattern changes. NanoGPT then supplies the one thing CIFAR cannot: matrices wide enough for the layer-rank term of Corollary 1 to be visible, which is where the two models of EF21-MuonSign separate.

Computing infrastructure. The experiments were not all run on the same machine. The synthetic study ran on one NVIDIA RTX A4000 (128-core AMD EPYC 7543 host, 472 GB RAM; Linux 6.12, Python 3.12, PyTorch 2.7.0, CUDA 12.8, driver 575.51.03); the language-modelling runs used a rented 8×H100 SXM node, specified in full in Appendix A.16. The centralized and federated CIFAR-10 runs were executed on single-GPU workstations. Runs written after the hardware stamp was introduced record the machine, commit and wall time in their own `metrics.json`, and `python3 -m common.hardware --scan results` prints the resulting table; the CIFAR runs predate it and record only the CUDA device index, so for those the machine is not recoverable from the released artefacts.

Randomness and seeds. Seeding is routed through a `seed_everything` procedure that seeds the Python `random` module, NumPy, and PyTorch (including all CUDA devices) and enables deterministic cuDNN kernels (`cudnn.deterministic = True`, `cudnn.benchmark = False`). The federated experiment of Table 2 reports five seeds (0–4) per method and the centralized experiment of Table 1 three (0–2), each quoted as mean ± one sample standard deviation across seeds; the nanoGPT runs of Table 3 and the weight-decay ablations use a single fixed seed (default 0), and Table 3 states the seed noise of the upstream reference so that the resolvable differences there are explicit. The synthetic study of Appendix A.12 averages over three draws of the problem itself (seeds 1337–1339) at a fixed $\mathbf{X}_0$ (seed 42), since every claim it makes concerns a *random* instance and a single draw cannot substantiate one, the alignment distribution least of all. The distinction matters for reading the tables: a difference smaller than the seed spread is not a result, and only the centralized and federated tables measure that spread. We report those spreads rather than a significance test, because at these seed counts no test could return anything. A paired Wilcoxon signed-rank test over n seeds has smallest attainable two-sided exact p-value 2^{1-n}, which is 0.0625 at the five federated seeds and 0.25 at the three centralized ones: even if every seed ordered two methods the same way, neither could reject at the 5% level. The comparisons we do claim are those separated by several standard deviations, for which the spread is the honest statistic; the differences we decline to claim are exactly those a test would have been asked to adjudicate. Learning-rate selection is performed once, at seed 0, on a validation split disjoint from the test set, and the selected rate is then reused unchanged for every seed.

Step-size schedules. Each experiment inherits the schedule conventional to its domain, and none of them is the constant rate our rates are proved for. The ResNet runs anneal cosinally to zero, the standard for this architecture and the premise of every accuracy we can be compared against; the nanoGPT runs keep record #40's stable-then-decay schedule, flat for the first 55% of steps and linear to $0.1\eta_0$ thereafter (Appendix A.16), because altering it would forfeit the reproduction that validates our port. The discrepancy is deliberate: matching the analysed step size would sacrifice comparability on both benchmarks and close only one of several gaps between the theory and the runs, the others being Newton–Schulz in place of an exact oracle, momentum, and normalization layers. One measurement is exempt: the growth-exponent diagnostic of Appendix A.17, which is run at a constant rate because under a decaying one the accumulated update saturates and the fit reports the schedule instead of the alignment.

Learning-rate selection. The only tuned hyperparameter is η_0. Every method with a norm-fixed step (the eight SIGN- and LMO-family rules of Appendix A.17, SignSGD included) is tuned and reported under one per-layer rule, the unit-gain rule (7), so that η_0 means the same thing across them, namely the per-step RMS gain, and no comparison within the families is confounded by the scaling convention. SGD and Adam have no norm-fixed step for the rule to act on and are run as practitioners run them, with one global rate. Because (7) is a heuristic rather than a theorem, the SIGN family is additionally re-tuned under the two competing conventions: one global rate, and μP's $1/\text{fan-in}$ (Yang et al. 2021). Appendix A.17 reports whether the ranking moves. Momentum is fixed at 0.9 and weight decay at 0 for the primary tables, the setting Mishra, Trivedi, and Kumar (2026)'s own sweep selects; the regularized case is an ablation (Appendix A.13 centralized, and below for the federated study). The auxiliary group, comprising biases, normalization parameters and the classifier head, is trained by AdamW at a fixed 10^{-3} for every method, so that the matrix rule is the sole quantity that varies.

In the centralized study, selection uses a fixed 45k/5k train/validation partition and validation accuracy alone, averaged over the last five epochs rather than read off a single one; **the test set is never consulted during tuning**. Each method receives the same number of configurations, drawn from a 1–2–5 lattice (three points per decade), so that tuned rates are exactly reproducible from the tables and every method searches the same grid. An optimum at an endpoint is treated as a failure rather than a result: the grid is widened along the lattice and the method re-run, up to four times. The selected η_0 is then retrained on the full 50k training set for 75 epochs under a cosine schedule, at three seeds, and we report the mean and standard deviation of the test accuracy over the last five epochs. Differences smaller than the seed spread are not claimed as results.

Tuning the federated study. Comparing sign placements is informative only if no placement is handicapped by its step size, so every method receives the same tuning budget: a five-point 1–2–5 lattice in η_0, ranked at a 400-round proxy horizon on a 5k validation split held out of the 50k *before* the client partition, with the grid widened and the method re-tuned whenever an optimum occurred at an endpoint. The reported runs then use the full 50k at the selected rate, so no test image is ever scored during selection. Two assumptions this protocol makes are checked rather than asserted: that the short proxy ranks rates the same way the full horizon does (re-running the top two rates of SignMuon and EF21-MuonSign at 2000 rounds returns the same winner in both cases), and that the shape dependence of the step is carried by the per-layer multiplier rather than by η_0. The second assumption is borne out: seven of the nine methods carrying a per-layer multiplier (the eight of Appendix A.17 plus the server-side-LMO Muon control) select the identical $\eta_0 = 0.05$ across both families, the remaining two (Muon at 0.1, SignSGD at 0.02) selecting a value one lattice step away. Weight decay is 0 in the reported table, matching the setting the theorems analyse; switching on a decoupled 5×10^{-4} moves the top three methods by at most 0.32 points and reorders nothing.

Accuracy and the threshold column. Alongside final accuracy we report the number of epochs (rounds, in the federated tables) to reach a fixed test accuracy. The two measure distinct quantities: with the methods spanning about a point and a half at 75 epochs (Table A5), final accuracy is close to the noise floor, whereas a threshold crossing on a monotonically rising curve separates the methods by factors rather than by tenths of a point, and is the analogue of the "steps to 3.35" column of Table 3.

Conventions with numerical consequences. Three implementation conventions can displace reported numbers and are recorded here. (i) The Muon LMO is computed in `bfloat16` (five Newton–Schulz steps) unless stated otherwise; for the methods that sign the LMO *output*, entries of $\mathrm{polar}(\cdot)$ near zero may flip at this precision, so that their trajectories carry a precision-dependent component (`-lmo-dtype float32` is available). (ii) In the federated runs, BatchNorm running statistics are never updated: local models are discarded each round and BN runs in inference mode during gradient accumulation, so the statistics stay at their initialization for the entire run, in training and evaluation alike. The result is a fixed normalization with learnable affine parameters, self-consistent between train and test, applied identically to every method. It is also one reason a channel may remain inactive across an entire local batch and so contribute an exactly zero row to the momentum. (iii) For EF21-MuonSign, training metrics are logged at the broadcast model $\mathbf{W}$ (where gradients must be evaluated) while validation and test metrics are evaluated at the server model $\mathbf{X}$ of (13), the iterate the guarantee bounds, except where a table states otherwise.

A.12 The smooth convex problem

On a deterministic L-smooth convex quadratic we measure the scalar the guarantees rest on: the alignment $\rho_t = \langle \nabla F(\mathbf{X}_t), \mathbf{D}_t \rangle / (\|\nabla F(\mathbf{X}_t)\|_F \|\mathbf{D}_t\|_F)$ between the gradient and the step. Theorems 1–3 construct instances driving it negative; on random instances the three methods they cover keep $\rho_t \geq 0.142$ throughout a tuned trajectory, so the construction is not one that random data reproduces. Only EF21-MuonSign becomes negative, on 0.95% of steps and to -0.026; it is also the one method here possessing a convergence guarantee (Appendix A.10), obtained without per-step descent.

The same experiment separates the two effects that a fixed-target iteration count confounds. Sign compression of an LMO step secures a lower accuracy floor, not a faster rate: SignMuon's floor lies a factor 1.74–1.80 below SignSGD's at every step size, and the two share $\|\mathbf{S}\|_F$ exactly, so the separation resides in the floor rather than in the step length.

Construction. To isolate the effect of matrix structure from stochastic noise and from the complexity of DNN architectures, we use a deterministic L-smooth convex quadratic:

$$F(\mathbf{X}) = \frac{1}{2}\|\mathbf{A}^{1/2}\mathbf{X}\mathbf{B}^{1/2}\|_F^2 = \frac{1}{2}\langle \mathbf{X}, \mathbf{A}\mathbf{X}\mathbf{B} \rangle \to \min_{\mathbf{X}\in\mathbb{R}^{m\times n}}, \tag{30}$$

with $\mathbf{A} \in \mathbb{S}^m_{++}, \mathbf{B} \in \mathbb{S}^n_{++}$ symmetric and $\mathbf{X}_0$ drawn entrywise from $\mathcal{N}(0, 0.01)$. Eigenvalues are sampled uniformly from $(0, 1)$ in a Haar-random eigenbasis, so the matrices are almost surely positive *definite* and the minimizer is $\mathbf{X}^\star = \mathbf{0}$ with $F^\star = 0$.

Two facts about this instance are exact rather than estimated, and both are used below. The Hessian of F is the Kronecker product $\mathbf{B} \otimes \mathbf{A}$, so its eigenvalues are the products $\lambda_i(\mathbf{A})\lambda_j(\mathbf{B})$: the Frobenius smoothness constant is $L = \max_{ij} \lambda_i(\mathbf{A})\lambda_j(\mathbf{B}) \leq 1$ and the strong-convexity constant is $\sigma = \min_{ij} \lambda_i(\mathbf{A})\lambda_j(\mathbf{B}) > 0$. The uniform draw leaves the resulting condition number L/σ uncontrolled (it is near 3.7×10^4 at the $m = n = 100$ every measurement in this subsection uses), so where conditioning is the variable we instead use log-spaced spectra with $L = 1$ and L/σ set exactly. And since $\nabla F(\mathbf{X}) = \mathbf{A}\mathbf{X}\mathbf{B}$ in closed form, the gradient can be evaluated at any point without an autograd graph, which permits the bidirectional method to be scored on its exact model $\mathbf{X}$ while its gradient is taken at the broadcast model $\mathbf{W}$, as its algorithm requires.

What "iterations to a target" measures. The natural criterion, fewest iterations to $F(\mathbf{X}) \leq 10^{-3}$ within $T_{\max} = 5000$ with learning rate and momentum tuned per method, proves not to measure a convergence rate, and reporting it as one is misleading. Every method here except SGD takes a *normalized* step: $\|\operatorname{sign}(\cdot)\|_F = \sqrt{mn}$ and $\|\operatorname{polar}(\cdot)\|_F = \sqrt{r}$ whatever the gradient does. A constant η therefore cannot converge; the iterate settles into a ball of radius $\eta\|\mathbf{S}\|_F$ and F plateaus. Write F_∞ and g_∞ for the settled values of $F(\mathbf{X}_t)$ and of $\|\nabla F(\mathbf{X}_t)\|_F$ on that plateau. Measured directly, $g_\infty \propto \eta$ for every method possessing a floor, $F_\infty \propto \eta^2$ for every such method but SignSGD, whose exponent is 1.33, and the iteration count is const/η. The tuner accordingly returns the largest η whose plateau falls under the target, and the resulting ranking is one of *accuracy floors*. The two effects disagree here. Over the seven step sizes at which both methods were run, SignMuon's g_∞ is 1.74 to 1.80 times below SignSGD's, a ratio flat in η as two floors of equal exponent must be, and its F_∞ between 6.7 and 71 times below, a ratio that is *not* flat, the two F-exponents differing. Tuned per budget, SignMuon holds the smaller $\min_t \|\nabla F\|_F$ at every horizon but $T = 250$, where SignSGD prevails by 2% (2.25 against 2.30×10^{-2}), a margin below what three draws resolve. We therefore report floor and rate separately, reading the descent lemma

$$F(\mathbf{X}_{t+1}) \leq F(\mathbf{X}_t) - \eta\langle \nabla F(\mathbf{X}_t), \mathbf{D}_t \rangle + \tfrac{\eta^2 L}{2}\|\mathbf{D}_t\|_F^2 \tag{31}$$

as the statement that separates them: the second term is the floor, the first is the rate.

Table A2 reports the criterion nonetheless, since it is the one the literature on these methods quotes. Read as a ranking of floors, it orders the six placements identically within both families: sign after the LMO is the least expensive, sign on both channels the most. Error feedback adds 10% on SignMuon and 17% on MuonSign, and removes 6% on MuonUSign, the one placement whose LMO already receives a compressed argument. Muon surpasses all six placements, and SGD (66) and Adam (85) surpass every normalized step by factors of three to ten, a quadratic with an exactly known Hessian being precisely the case for which a scaled gradient step is designed. Figure A2 plots the trajectories behind those counts, and shows the plateau that makes the criterion a ranking of floors.

Alignment. Equation (31) makes progress contingent on a single scalar, the normalized alignment between the gradient and the step actually taken,

$$\rho_t := \frac{\langle \nabla F(\mathbf{X}_t), \mathbf{D}_t \rangle}{\|\nabla F(\mathbf{X}_t)\|_F \, \|\mathbf{D}_t\|_F} \in [-1, 1]. \tag{32}$$

Theorems 1–3 are constructions that drive ρ_t negative. Table A3 reports its distribution along the tuned trajectory on random instances, which is the empirical counterpart of those theorems and the one measurement here that is about the methods rather

Algorithm	**iters**	**best** F	$\min \|\nabla F\|_F$	**tuned** (η, μ)
SignMuon	364	$6.9 \cdot 10^{-4}$	$2.4 \cdot 10^{-2}$	$(1.0 \cdot 10^{-3}$, 0)
MuonUSign	556	$7.1 \cdot 10^{-4}$	$2.3 \cdot 10^{-2}$	$(1.0 \cdot 10^{-2}$, 0.5)
MuonSign	595	$4.8 \cdot 10^{-4}$	$1.9 \cdot 10^{-2}$	$(6.8 \cdot 10^{-4}$, 0.5)
EF21-SignMuon	401	$5.4 \cdot 10^{-4}$	$2.1 \cdot 10^{-2}$	$(6.8 \cdot 10^{-3}$, 0)
EF21-MuonUSign	521	$5.3 \cdot 10^{-4}$	$1.8 \cdot 10^{-2}$	$(6.8 \cdot 10^{-3}$, 0)
EF21-MuonSign	695	$3.8 \cdot 10^{-4}$	$1.7 \cdot 10^{-2}$	$(4.6 \cdot 10^{-3}$, 0)
Muon	267	$4.7 \cdot 10^{-4}$	$2.0 \cdot 10^{-2}$	$(1.0 \cdot 10^{-2}$, 0.5)
SignSGD	486	$4.3 \cdot 10^{-4}$	$2.2 \cdot 10^{-2}$	$(6.8 \cdot 10^{-4}$, 0.5)
SGD	66	$1.9 \cdot 10^{-8}$	$1.6 \cdot 10^{-6}$	(2.15, 0.5)
Adam	85	$9.2 \cdot 10^{-6}$	$6.5 \cdot 10^{-4}$	$(6.8 \cdot 10^{-2}$, —)

Table A2: Fixed-target criterion under the current protocol: fewest iterations to $F \leq 10^{-3}$ within $T_{\max} = 5000$ at $m = n = 100$, over three problem draws, $(\eta, \mu, \text{schedule})$ tuned on five-decade logarithmic grids. A constant schedule is selected for every method. Best F and $\min \|\nabla F\|_F$ are minima over all 5000 iterations rather than values at the crossing, so they report the floor the method settles into.

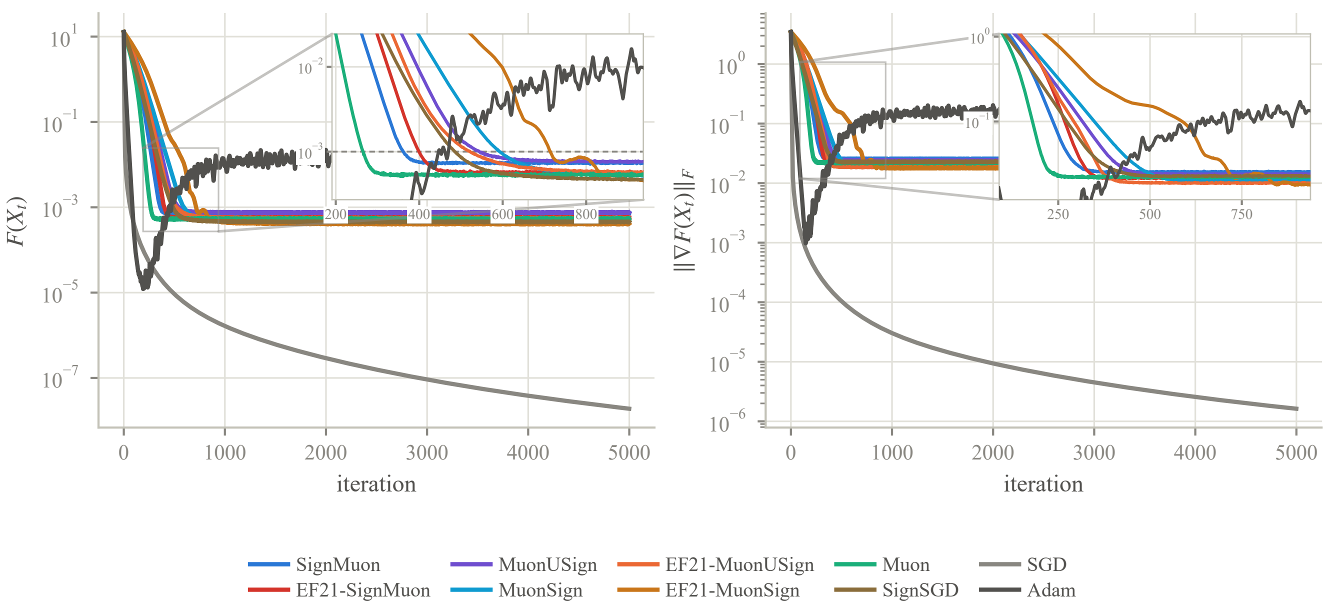

Figure A2: Trajectories at the optima of Table A2, mean over three problem draws. Every normalized method reaches a plateau within a few hundred iterations and remains there, which renders the crossing time a ranking of floors. SGD, whose step vanishes with the gradient, is the only curve still descending at $T_{\max}$; Adam oscillates about its floor rather than settling onto it. The insets magnify the window in which the methods arrive, which over the full axis is a fold of curves in the first sixth of the range: the dashed line is the target $F \leq 10^{-3}$, and the order in which the curves cut it is the iteration count of Table A2, Muon first and EF21-MuonSign last. The window runs to a little past the last crossing and is scaled to the eight norm-fixed methods. One caveat on reading a count off it: the tabulated counts are means of the per-draw crossing times, while a plotted curve is the geometric mean of the trajectories, and for the slowest method the two differ by enough to see.

than about the tuning protocol. Three closed forms anchor it: $\rho = 1$ for SGD, $\rho = \|\mathbf{G}\|_1/(\|\mathbf{G}\|_F\sqrt{mn})$ for SignSGD, and $\rho = \|\mathbf{G}\|_*/(\|\mathbf{G}\|_F\sqrt{r})$ for Muon. The six sign-around-the-LMO methods admit none, which is the subject of this paper.

Floor, budget and stability. Three further measurements, each testing a quantity the theory predicts in closed form. (i) The floor: the fitted slope of $\log g_\infty$ against $\log \eta$, against the 1 that Equation (31) predicts. Balancing the two terms of (31) at $\langle \nabla F, \mathbf{D} \rangle = \rho \|\nabla F\|_F \|\mathbf{D}\|_F$ gives $\eta\rho\|\nabla F\|_F\|\mathbf{D}\|_F = \eta^2 L \|\mathbf{D}\|_F^2/2$, hence $g_\infty = \eta\, L\|\mathbf{S}\|_F/(2\rho)$: slope 1 with that coefficient. Since SignMuon and SignSGD share $\|\mathbf{S}\|_F = \sqrt{mn}$ exactly, any separation between their floors is attributable to ρ alone. (ii) The budget exponent: tuning $(\eta, \mu, \text{schedule})$ separately at each horizon T and fitting err $\propto T^{-p}$, $\eta^\star \propto T^{-q}$, where err $:= \min_{t \leq T} \|\nabla F(\mathbf{X}_t)\|_*^2$ is the *squared* dual norm our theorems bound. Reporting the norm itself halves p and leaves q unchanged: q describes the tuned step size, which is indifferent to the power of the error we elect to plot. The nonconvex

Algorithm	$\min_t \rho_t$	median ρ_t	mean ρ_t	% of steps with $\rho_t < 0$	closed form	tuned (η, μ)
SignMuon	0.208	0.381	0.399	0.00%	—	$(2.2 \cdot 10^{-4}, 0)$
MuonUSign	0.161	0.357	0.378	0.00%	—	$(3.2 \cdot 10^{-3}, 0)$
MuonSign	0.142	0.360	0.364	0.00%	—	$(2.2 \cdot 10^{-4}, 0)$
EF21-SignMuon	0.333	0.659	0.573	0.00%	—	$(1.5 \cdot 10^{-3}, 0)$
EF21-MuonUSign	0.346	0.558	0.553	0.00%	—	$(1.5 \cdot 10^{-3}, 0)$
EF21-MuonSign	−0.026	0.489	0.419	0.95%	—	$(1.5 \cdot 10^{-3}, 0)$
Muon	0.454	0.666	0.633	0.00%	0.695	$(1.5 \cdot 10^{-3}, 0)$
SignSGD	0.174	0.503	0.499	0.00%	0.794	$(2.2 \cdot 10^{-4}, 0.5)$
SGD	—	—	—	—	1	$(3.2, 0.95)$

Table A3: Alignment ρ_t of Equation (32) along the tuned trajectory, over three problem draws at $m = n = 100$. The closed form is evaluated at $\mathbf{X}_0$ without momentum and is comparable only to rows whose tuned μ is 0; with momentum the step is built from $\mathbf{M}_t$ and ρ_t falls accordingly. The six sign-around-the-LMO methods admit no closed form. SGD's is 1, but its step is not instrumented, so the distribution is left empty rather than filled with the constant; Adam is omitted, its step lying outside the descent lemma. Every method of Theorems 1–3 remains clear of zero. The single negative excursion is EF21-MuonSign's, which Appendix A.10 proves convergent through the EF21 estimator rather than through per-step descent.

	floor		budget				stability
Algorithm	$\mathrm{d}\log g_\infty / \mathrm{d}\log\eta$	R^2	p	R^2	q	R^2	$\eta_{\max}\|\mathbf{S}\|_F$
SignMuon	1.000	1.000	1.88	0.996	0.39	0.94	13.4
MuonUSign	1.000	1.000	1.86	0.995	0.39	0.94	14.9
MuonSign	1.000	1.000	1.93	0.993	0.44	0.94	11.8
EF21-SignMuon	1.000	1.000	2.07	1.000	0.55	1.00	18.9
EF21-MuonUSign	1.000	1.000	2.04	1.000	0.55	1.00	14.9
EF21-MuonSign	1.000	1.000	2.14	1.000	0.55	1.00	14.9
Muon	1.000	1.000	2.03	0.999	0.55	1.00	17.6
SignSGD	0.989	1.000	1.76	0.990	0.39	0.94	11.1
SGD	*no floor*		8.97	0.985	−0.11	0.50	2.06†
Adam	1.000	1.000	2.65	0.915	−0.66	0.62	> 100‡
predicted	1		1/2 or 1		1/2 or 1		$2/L$

Table A4: Floor exponent, budget exponents and stability edge against the predicted values, at $m = n = 100$ over three problem draws. The floor exponent attains its prediction exactly for eight of the nine methods possessing a floor, and to within 1.1% for the ninth (SignSGD). The eight norm-fixed methods tune their step size as the nonconvex bound prescribes, q falling in $[0.39, 0.55]$ about the predicted $1/2$ at $R^2 \geq 0.94$, yet converge faster than either bound admits: $p \approx 2$ at $R^2 \geq 0.99$, twice the strongly convex value, a quadratic being easier than the worst case of its smoothness class. SGD possesses no floor, its step vanishing with the gradient, so its floor columns are empty by construction and its $\eta_{\max}$ is the control reproducing $2/L = 2.019$; neither baseline admits a power law in $\eta^\star$. † SGD has no $\|\mathbf{S}\|_F$; the entry is $\eta_{\max}$, 1.02 times $2/L$. ‡ The search reached its ceiling without encountering an edge: Adam's step is bounded by roughly η irrespective of the gradient, so it oscillates rather than diverges.

bound our theorems prove gives $p = q = 1/2$; a strongly convex problem gives $p = q = 1$, because the rate term contracts geometrically and the error is floor-limited. The fit therefore reports which regime the instance occupies, and this one, having $\sigma > 0$, need not occupy the regime the theorems describe. It reports both at once: the eight norm-fixed methods tune $\eta^\star$ as the nonconvex bound prescribes, q scattering about $1/2$, whereas the error they attain falls at $p \approx 2$, twice the strongly convex value and four times the nonconvex one. What the pair describes is a step size chosen for the worst case of the smoothness class, applied to an instance far easier than that worst case. The schedule is tuned per method rather than imposed, since nothing obliges the eight methods to prefer the same one. (iii) The stability edge: the largest stable η, with SGD as a control, its value being required to reproduce the textbook $2/L$. Reported as a step length $\eta_{\max}\|\mathbf{S}\|_F$, this would be family-independent were the operative trust region the Frobenius ball; the spread measures how far the Frobenius bound stands from the geometry in which each step actually resides.

Conditioning. Conditioning governs the dynamics of a quadratic, and the construction above leaves it to chance, so the right panel of Figure A3 sweeps κ over five decades, from 10 to 10^6, at fixed $L = 1$. The eight norm-fixed methods are insensitive to it, with $\mathrm{d}\log\|\nabla F\| / \mathrm{d}\log\kappa$ between -0.096 (Muon) and -0.17 (EF21-SignMuon), so that over the entire sweep the attained $\|\nabla F\|$ moves by a factor of 3.2 (Muon) to 8.3 (EF21-MuonSign). The sign is mildly negative because at fixed L a larger κ

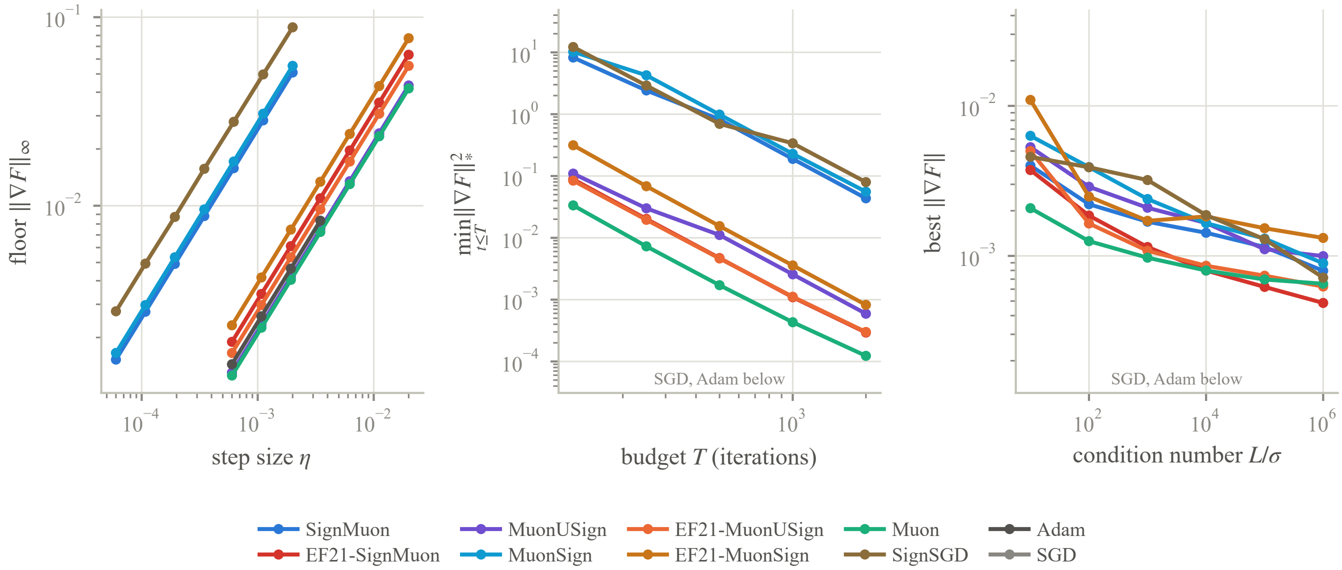

Figure A3: Log–log fits behind Table A4, at $m = n = 100$ over three problem draws. *Left*, the accuracy floor of a constant step: the lines are parallel because every exponent is 1, and the two clusters are the two step lengths, the three methods with $\|\mathbf{S}\|_F = \sqrt{mn}$ on the left and the five with $\sqrt{r}$ on the right, each swept over the window its length calls for. *Centre*, best error at each tuned budget, in the norm dual to each method's own LMO ball; the two bands are the two duals, ℓ_1 above and nuclear below, so the curves are comparable within a family but not across families. *Right*, best $\|\nabla F\|$ within the budget against L/σ, tuned at each point; across the five decades swept, the eight norm-fixed methods vary by a factor of 3.2 to 8.3, against the sixteen orders of magnitude spanned by SGD and Adam. In the last two panels the vertical range is set by those eight methods, and SGD and Adam leave the frame below: on a quadratic with an exactly known Hessian a scaled gradient step converges to machine precision, fifteen decades below anything with a norm-fixed step, and on a shared range the eight would occupy a single line.

entails a smaller σ, hence a flatter landscape to occupy rather than a harder one to descend; their floors are fixed by $\eta\|\mathbf{S}\|_F$, which is independent of the spectrum. SGD and Adam are sensitive to it, jointly spanning sixteen orders of magnitude across the sweep, which is why the panel is scaled to the eight and leaves those two below it. SGD's fitted slope of 3.6 is not an exponent: its first three points lie at or below 10^{-20}, which on this problem constitutes exact convergence rather than a measurement.

Protocol and reproduction. Every measurement above is at $m = n = 100$ over three problem draws (seeds 1337–1339, $\mathbf{X}_0$ from seed 42), with the LMO taken in `bfloat16` as in the network experiments, and is reported as the geometric mean over the draws for the error metrics and the arithmetic mean for iteration counts.

Learning rates are searched on logarithmic grids spanning five decades, one per step-norm family, each ending past the largest stability edge measured for that family so that no stable step size falls outside the search. A one-decade linear grid, which an earlier version of this experiment used, is narrow enough to censor an optimum at a grid boundary, giving an upper bound rather than a tuned value; the search flags any optimum at an edge. No learning rate reported above is flagged. The windows are derived from $\|\mathbf{S}\|_F$ rather than from the method name, which matters for MuonUSign and EF21-SignMuon: both take a step of length $\sqrt{r}$ despite the SIGN in their names, and under a window assigned by name both were searched a decade below the range their step length calls for.

The remaining flags are on momentum, at the top of its grid (0.99), and belong to SGD and SignSGD at the largest condition numbers: an ill-conditioned quadratic asks for heavy momentum, so those two rows of the κ sweep bound the dependence from one side rather than measuring it. The other eight methods select 0.5 or less at every κ, and the three carrying an error-feedback estimator select 0 throughout.

Every number in this subsection comes from a single scripted run of the released benchmark, which records each stage together with the commit, GPU and wall time it ran under.

A.13 Centralized training results

Table A5 reports the centralized CIFAR-10 results on ResNet-18 in full: the selected η_0, the final train and test accuracies, the epoch count to the 90% threshold, and the per-epoch cost. Figure 2 starts at epoch 25, where the methods are a few points apart rather than twenty. Every run uses batch 128, momentum 0.9, auxiliary rate 10^{-3}, and zero weight decay, so that the matrix rule and η_0 are the only quantities that vary. Figure A4 gives the accuracy curves from epoch 1, Figure A5 the training loss, and Figure A6 the learning-rate sweep behind the selection.

Dataset	Optimizer	Epochs	η_0	Train Acc	Test Acc	Ep. to 90%	s/epoch
CIFAR-10	EF21-SignMuon	75	0.01	100.00	94.64 ± 0.24	8.7	14.8
	SignMuon		0.02	100.00	94.53 ± 0.22	8.3	14.4
	Muon		0.05	99.99	94.49 ± 0.09	7.7	14.4
	EF21-MuonSign		0.01	99.99	94.22 ± 0.13	10.3	15.3
	EF21-MuonUSign		0.2	99.99	94.15 ± 0.16	8.7	14.9
	MuonUSign		0.02	99.99	93.90 ± 0.08	10.0	14.4
	MuonSign		0.005	99.98	93.65 ± 0.21	15.0	14.5
	Adam		0.0005	99.98	93.35 ± 0.15	18.8	12.7
	SignSGD		0.002	99.97	93.27 ± 0.16	19.0	12.1
	SGD		0.02	99.95	93.13 ± 0.07	21.7	12.2

Table A5: CIFAR-10: centralized learning on ResNet-18, three seeds. Test accuracy is the mean $\pm$ standard deviation of the last five epochs; train accuracy, "Ep. to 90%" and "s/epoch" are means over the same three seeds. η_0 was selected on a held-out 5k validation split at 15 epochs and then retrained on the full 50k set, as described in Appendix A.11. Every method fits the training set to within 0.05 points of the others, so the column separates nothing on its own; the differences in the test column are not differences in how far training got.

Horizon stability. Selection was carried out at 15 epochs and the reported runs are 75, so the tuning horizon and the reporting horizon differ. Re-tuning the top three rates directly at 75 epochs moves the winner for two methods: SignMuon from $\eta_0 = 0.02$ to 0.2 and Muon from 0.05 to 0.01. Both moves are small in accuracy, SignMuon reaching 94.53 ± 0.22 at 0.02 against 94.29 at 0.2 (one seed), and the sweep in Figure A6 is flat across that range for every LMO method. The short-horizon selection is nonetheless provisional in the strict sense, and we report it as such rather than presenting the 15-epoch grid as though it settled the 75-epoch question.

Weight decay. The primary table is unregularized, the setting the theorems analyse and the one Mishra, Trivedi, and Kumar (2026)'s sweep selects. Repeating the top three at the same η_0 with decoupled decay 5×10^{-4} (seed 0, decay not re-tuned) leaves the ordering intact: EF21-SignMuon 94.52% against 94.55% undecayed, Muon 94.33% against 94.53%, SignMuon 94.35% against 94.75%. Decay is applied decoupled, $\mathbf{X} \mathrel{*}= 1 - \eta\lambda$, so the LMO sees the true gradient; the coupled convention would only rotate the direction, since every step here is scale-invariant.

A.14 Communication accounting

Table 2 quotes mean bits per parameter per round. This section states precisely what is counted, since the headline "32×" of the sign-compression literature is an idealization eroded by three separate effects, only one of which is customarily acknowledged.

Write P_{mat} for the number of matrix parameters, P_{aux} for the auxiliary group (biases, BatchNorm affine parameters, the classifier head), L for the number of matrix layers, and $P = P_{\text{mat}} + P_{\text{aux}}$. On CNN2, $P_{\text{mat}} = 762{,}560$, $P_{\text{aux}} = 2{,}146$ and $L = 3$. All figures below are *per client per round*: the uplink is what one client sends, the downlink what the server sends to one client.

(i) The uplink alphabet. The randomized sign of Section 4 renders every transmitted symbol a genuine bit, so the uplink costs one bit per parameter with no entropy coding required to realize it.

(ii) The auxiliary group is never compressed. It travels at full precision in both directions for every method, so a "one-bit" channel actually costs

$$\frac{1 \cdot P_{\text{mat}} + 32\, P_{\text{aux}}}{P} = 1.087 \text{ bits},$$

a 29.4× reduction rather than 32×. On CNN2 the group is 0.28% of the parameters; on an architecture with a large embedding or head it would dominate this table, which is why the quantity is computed per model rather than quoted.

(iii) Error feedback carries one scale per layer. Both EF21 channels transmit the pair $(\mathbf{s}_t, \alpha_t)$ with one full-precision α_t per matrix layer: on the uplink for every EF21 method, and again on the downlink for EF21-MuonSign. That is $32L$ bits, adding $32L/P \approx 1.3 \times 10^{-4}$ bits per parameter here. That lies four decimal places in, and it is reported for completeness rather than because it alters a conclusion: it is the difference between "one bit per parameter" and "one bit per parameter, plus a constant".

(iv) Which methods compress the downlink. The criterion is not whether the method applies a compressor but whether the object the server must distribute is already ± 1-valued. Three cases qualify: the majority vote $\mathbf{s}_t^{\text{agg}}$ itself (SignMuon, SignSGD), a signed LMO output (MuonSign), and a primal error-feedback residual (EF21-MuonSign). In the first of these the server broadcasts the vote rather than the model and each client applies the step to its own replica; replicas start from a common $\mathbf{X}_0$ and receive identical updates, so they never drift. The remaining three methods must distribute a dense server-side quantity,

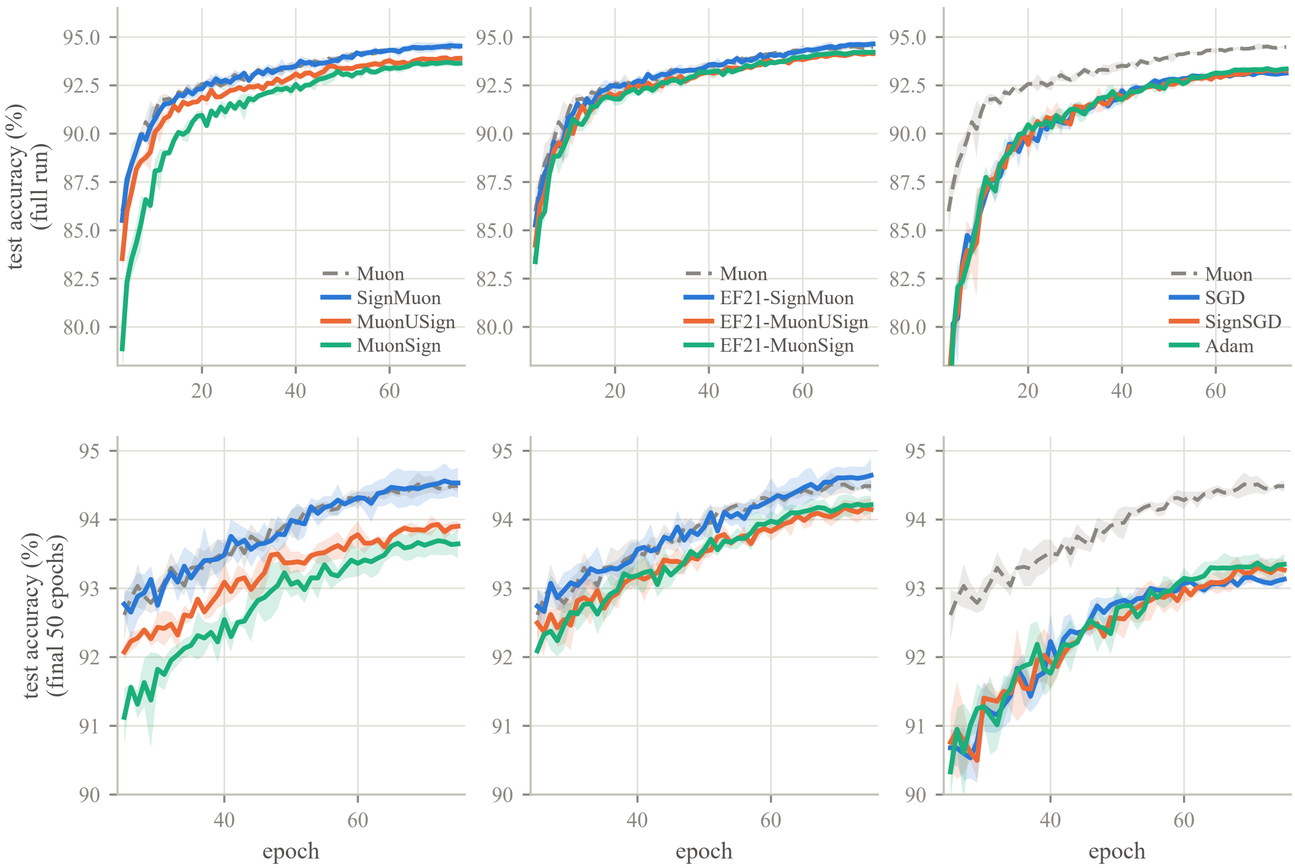

Figure A4: Test accuracy over the whole run (top row) and from epoch 25 (bottom row), the same data at two scales, grouped as in Figure 2. Bands are ± 1 standard deviation over three seeds; Muon is the gray dashed reference in every panel. Series are labelled at the right of each panel.

Method	**Up (bits)**	**Down (bits)**	**Up**	**Down**
Muon, MuonServer	32	32	1.0×	1.0×
SGD, Adam	32	32	1.0×	1.0×
SignMuon	1.0870	1.0870	29.4×	29.4×
SignSGD	1.0870	1.0870	29.4×	29.4×
MuonSign	1.0870	1.0870	29.4×	29.4×
EF21-MuonSign	1.0871	1.0871	29.4×	29.4×
MuonUSign	1.0870	32	29.4×	1.0×
EF21-SignMuon	1.0871	32	29.4×	1.0×
EF21-MuonUSign	1.0871	32	29.4×	1.0×

Table A6: Communication per client per round on CNN2, under the randomized-zero convention. Bits are per model parameter; reductions are against a 32-bit baseline. The fourth decimal separates the error-feedback methods from the rest: it is the per-layer scale of point (iii). Groups: uncompressed references, one bit in both directions, one bit on the uplink only.

namely $\mathrm{polar}(\cdot)$ of the aggregate for MuonUSign and EF21-MuonUSign and a scaled average of signs for EF21-SignMuon, and therefore transmit it at full precision.

Table A6 collects the four effects into the per-round cost of each method, and is the source of the Up and Down columns of Table 2. It is computed by `federated.algorithms.communication_bits`, which takes the measured zero rate of the run it describes, so a run made under the legacy ternary convention reports its own higher figure rather than the idealized one.

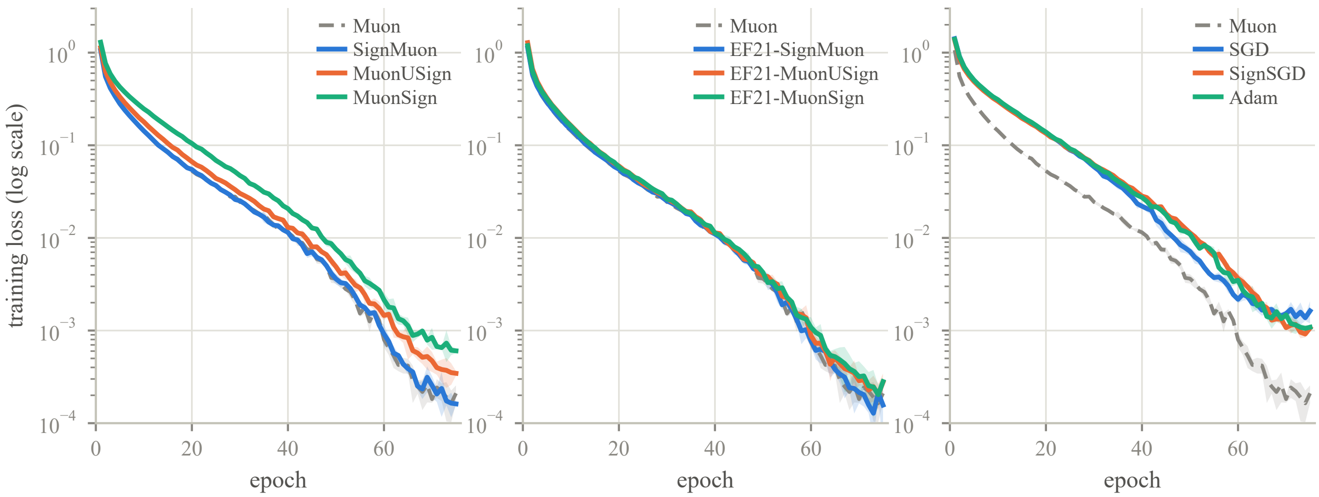

Figure A5: Training loss (log scale), grouped as in Figure 2 and over the same runs. Within a panel the methods are hard to separate (every one of them falls to 10^{-3} or below, and they differ mainly in how early), which is why the body figure carries the accuracy alone. What the loss does show is the gap between families: in the right panel Muon reaches a given loss some fifteen epochs ahead of SGD, SignSGD and Adam and ends nearly an order of magnitude lower, while in the left panel SignMuon tracks Muon and the sign-before and both-sides orderings trail it, in the same order as their accuracies.

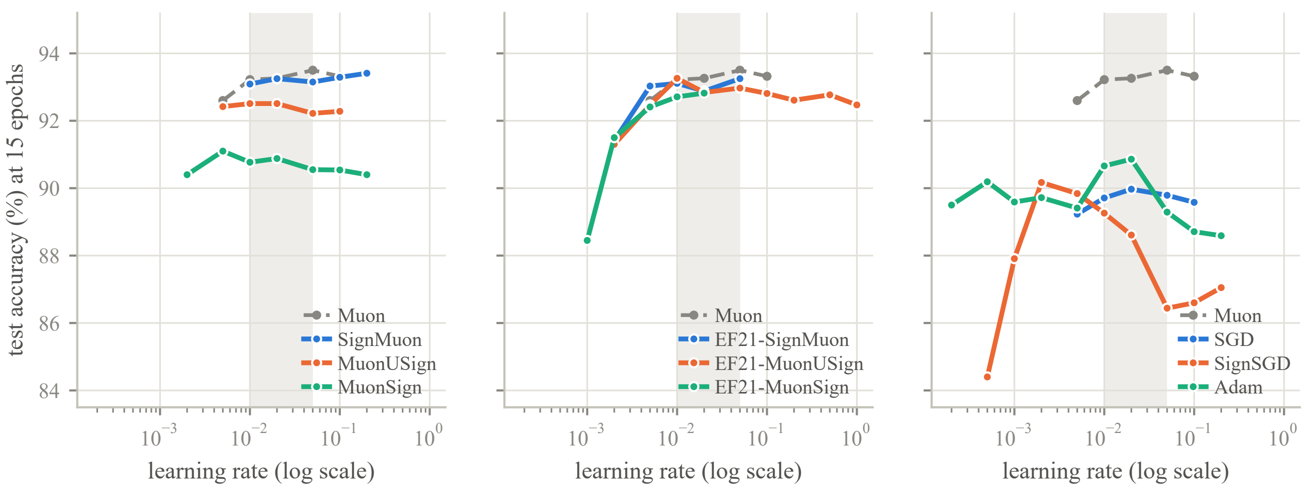

Figure A6: Learning-rate sweep at 15 epochs on the tuning split, grouped as in Figure 2, with Muon repeated as the gray dashed reference. The shaded band marks $\eta_0 \in [0.01, 0.05]$, the range every method except EF21-MuonSign was swept over and therefore the only range-matched comparison: within it the LMO methods vary by 0.16–0.42 points (Muon itself, 0.28, is no flatter than the sign variants), while Adam varies by 1.57 and SignSGD by 2.82. The robustness belongs to the LMO, and taking the sign of its output does not cost it.

A.15 Federated training results

Figure A7 (together with Table 2 in the main text) reports the comparison of optimizers at $N = 11$ clients. The figure additionally shows Muon with a server-side LMO, which is not a communication-efficient method but isolates whether moving the oracle off the clients carries a penalty on its own; it does not ($85.74 \pm 0.11\%$ against Muon's $85.99 \pm 0.26\%$), so the gaps in Table 2 are attributable to compression and sign placement rather than to where the oracle runs.

A.16 Language-modelling details

Setup. Upstream modded-nanoGPT record #40 (2025-10-04), the last record before NorMuon and hence the last whose hidden-matrix optimizer is a clean, separable momentum $\to$ LMO $\to$ step Muon, so our variants inject exactly at the LMO.

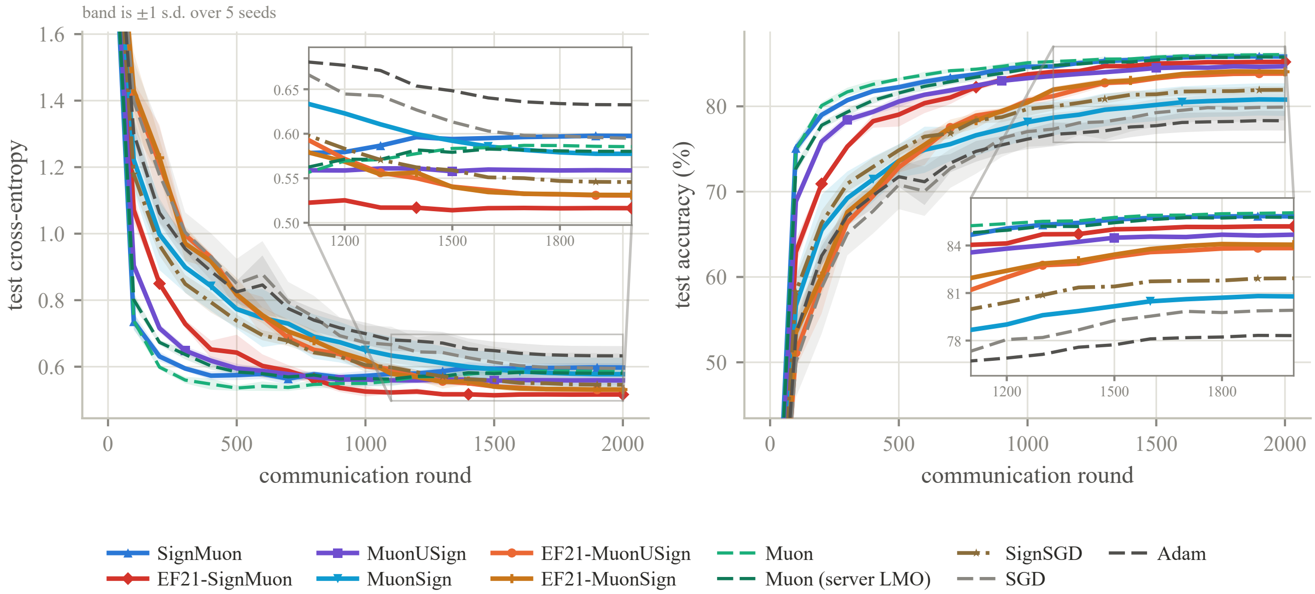

Figure A7: CIFAR-10 federated learning on CNN2 at $N = 11$ clients and batch 192 per client: test cross-entropy (left) and test accuracy (right) against communication round. Each curve is the mean over five seeds and the band is ± 1 sample standard deviation across them; all methods are evaluated on the exact server model $\mathbf{X}_t$. Both axes are clipped to exclude round 0, which is the untrained model and identical for every method. Line style carries the method's role, solid for the one-bit methods, dashed for the uncompressed references and dash-dot for SignSGD, so that hue is not the only channel separating eleven curves; markers are spaced apart per method for the same reason. The insets magnify the final 45% of rounds, where every number in Table 2 is read and where the curves lie within a line width of one another; bands are omitted inside them, since at that zoom they overlap into a single wash. Eleven methods are drawn, one more than Table 2, the extra being the server-side-LMO Muon control.

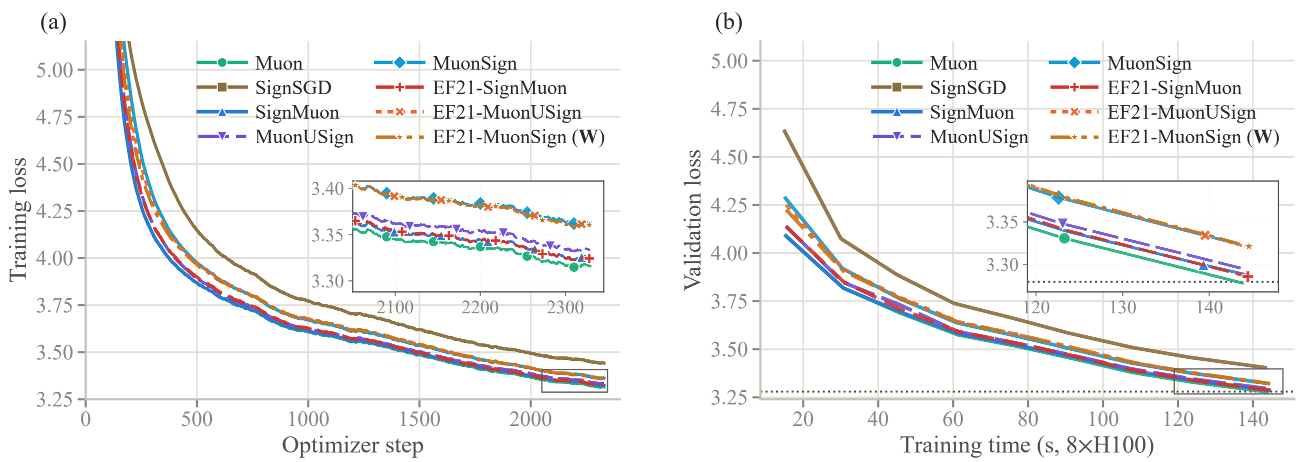

Figure A8: NanoGPT, supporting curves. (a) Training loss against optimizer step (EMA-smoothed; the logged quantity is a per-rank sum over 32,768 tokens, divided out here). (b) Validation loss against the speedrun clock, which excludes validation and compilation. Insets magnify the boxed tails, as in Figure 3. Marker positions are staggered in both figures so that curves lying within a line width of one another can still be followed individually. The ordering is the same on both axes: no method incurs a measurable wall-clock premium for its compressor or its error-feedback buffers. Note in (a) that EF21-MuonSign's training loss, taken at $\mathbf{W}$, exhibits no degradation throughout, which is what localizes its validation gap to the tracking of $\mathbf{X}$ rather than to training.

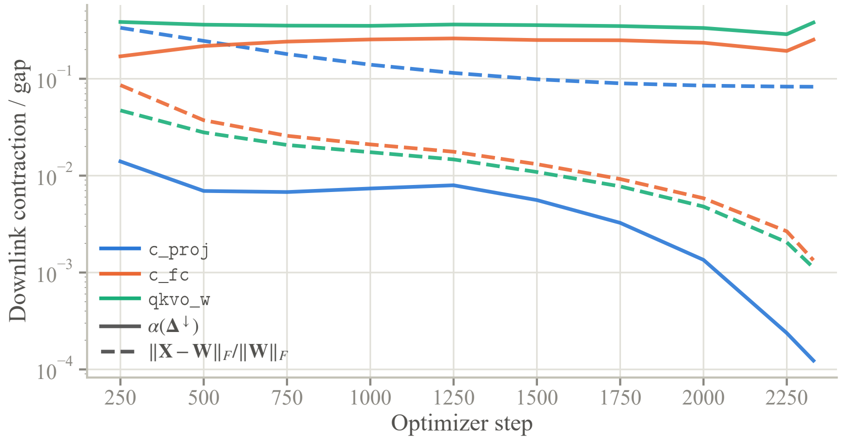

Figure A9: The downlink measurement for EF21-MuonSign, per layer type: the contraction $\alpha(\mathbf{\Delta}^{\downarrow})$ the scaled sign actually achieves (solid) and the resulting server/broadcast gap $\|\mathbf{X}_t - \mathbf{W}_t\|_F/\|\mathbf{W}_t\|_F$ (dashed). On the zero-initialized `c_proj` the contraction is two orders of magnitude below the rest from step 250 on and falls a further two by the end, and the gap stops decreasing near 10^{-1}; every other layer type contracts at $\Theta(10^{-1})$ and closes its gap by 6–64× over the run, ending below 10^{-2}. This is Remark 3 rendered as a measurement.

Model: 12 layers, model dimension 768, 6 heads of dimension 128, vocabulary 50,257; hidden matrices are 768×3072 (the merged $\mathbf{Q}/\mathbf{K}/\mathbf{V}/\mathbf{O}$ weight is used as four 768×768 blocks, and both the LMO and the compressor scale are applied per block). Data: FineWeb10B, the 10B-token sample of FineWeb (Penedo et al. 2024) that the speedrun repository prepares and tokenizes; 262,144 tokens per step, 2330 steps (= 611M tokens), validation on the fixed 10,485,760-token split. Hardware: one rented 8×H100 SXM node (80 GB per GPU; dual Xeon Platinum host, 224 vCPU, 2 TB RAM, PCIe 5.0 ×16, NVMe scratch), running the PyTorch NGC container under CUDA 13; one process per GPU. Gradients are averaged by `reduce_scatter` so the owning rank runs the centralized update and `all_gather` returns the parameter, i.e. the compression is a property of the update rule, as in the centralized algorithms we analyze.

Hyperparameters. Matrix/gate optimizer: $\eta_0 = 0.06$ (LMO family) or 0.03 (SIGN family), per-layer scaled by the unit-gain rule (Appendix A.17); Nesterov momentum $\mu = 0.95$, warmed up linearly from 0.85 over the first 300 steps and cooled back to 0.85 over the last 50; weight decay 0 (the record's own value); η constant then linearly cooled to $0.1\eta_0$ over the final 45% of the run; LMO by 5 Polar-Express iterations. Auxiliary parameters (embeddings, scalars, head) use the record's distributed Adam unchanged: $\eta = 0.008$, $\beta = (0.65, 0.95)$, $\varepsilon = 10^{-8}$, no weight decay, per-parameter multipliers 75 on embeddings and 5 on scalars, stepped every other iteration. Nothing above was tuned by us: the LMO family runs at the record's own η_0, and every value outside the matrix optimizer is the record's. Wall-clock varies by at most 1.2% across all eight methods (61.4–62.1 ms/step).

Supporting curves. Figure A8 gives the two views Figure 3 omits: training loss against optimizer step, and validation loss against the speedrun clock. The second is what supports the claim of equal wall-clock in Section 5.3; the first shows that EF21-MuonSign trains normally at $\mathbf{W}$, which is what places its validation gap in the tracking of $\mathbf{X}$ rather than in training. Figure A9 then measures that tracking directly, per layer type.

Compressor diagnostics. Table A7 reports, per layer type at the final step, the contraction each scaled sign achieves, $\alpha(\mathbf{\Delta}) = \|\mathbf{\Delta}\|_1^2/(d\|\mathbf{\Delta}\|_F^2)$, and the relative estimator lag. Three observations merit separate comment. *(i)* Every uplink is well-contractive, $\alpha \in [0.32, 0.64]$ against the isotropic $2/\pi = 0.637$, and EF21-SignMuon's is uniformly the best because its target is orthogonal; yet the uplink *lag* is large (0.36–0.82), so that the estimator remains far from its target at every step and the methods nonetheless train well, the LMO being scale-invariant and requiring only the direction. *(ii)* The single anomaly in the table is the `c_proj` downlink, $\alpha = 1.2 \times 10^{-4}$, four orders of magnitude below its uplink on the very same layer. Since both compressors are the same operator, the difference is a property of the residual, not of the compressor: the uplink residual is refreshed by an exogenous stochastic gradient each round, the downlink residual is generated by the compressor's own recursion (Remark 3). *(iii)* The resulting validation loss at the server model $\mathbf{X}$ is 4.20 through step 750, rises to 5.52 by step 1500, and then holds there (5.51–5.58 to the end of the run): a persistent offset above the broadcast model, not a divergence.

Where the two models part. The `c_proj` anomaly of item *(ii)* is what separates EF21-MuonSign's two models: on that layer the gap $\|\mathbf{X}_t - \mathbf{W}_t\|_F/\|\mathbf{W}_t\|_F$ stops decreasing at 8% against $<1\%$ for every other layer, and that one layer accounts for the entire ≈2.2-nat offset of item *(iii)*. The reason is the initialization. A layer built from zero receives maximally correlated updates, so its downlink residual concentrates, and a compressor that moves every coordinate by $\mathrm{mean}\,|\mathbf{\Delta}^{\downarrow}|$ cannot catch the coordinates

Layer type	uplink α	uplink lag	downlink α	downlink gap
EF21-SignMuon				
`qkvo_w` ($\times 10$)	0.640	0.66	–	–
`c_fc` ($\times 11$)	0.633	0.62	–	–
`c_proj` ($\times 11$)	0.634	0.62	–	–
EF21-MuonUSign				
`qkvo_w`	0.374	0.81	–	–
`c_fc`	0.323	0.82	–	–
`c_proj`	0.597	0.62	–	–
EF21-MuonSign				
`qkvo_w`	0.406	0.79	0.380	0.0011
`c_fc`	0.355	0.78	0.253	0.0013
`c_proj`	0.596	0.61	$\mathbf{1.2{\times}10^{-4}}$	$\mathbf{0.083}$

Table A7: Compressor diagnostics at step 2330, medians over the identical layers of each type. α is the contraction the scaled sign achieves on that round's residual ($2/\pi$ for an isotropic residual, $1/d$ in the worst case); "lag" is $\|\text{target} - \text{estimator}\|_F/\|\text{target}\|_F$; "gap" is $\|\mathbf{X}_t - \mathbf{W}_t\|_F/\|\mathbf{W}_t\|_F$. Downlink columns are empty for the two methods that broadcast exactly. Gates are omitted (they are 6×12 and 1×12).

driven hardest. This is the mechanism of Remark 3 rather than a tuning failure. Lowering η_0 does not repair it, because the admissible step size would have to shrink by a further $\sqrt{r}$ in the layer rank (Remark 4); only a downlink compressor contractive in the *spectral* norm would.

A.17 Per-Layer Step Sizes: the Unit-Gain Rule

This appendix derives the heuristic (7) stated in the main text, selects its one free exponent by measurement, and records what it does and does not deliver (including the two methods it does not act on at all, SGD and Adam, which have no norm-fixed step and are run at one global rate throughout).

Our counterexamples, like the analysis of Mishra, Trivedi, and Kumar (2026), concern a single matrix, where one scalar step size suffices. A network has layers of very different shapes, and the methods of this paper produce step matrices from *two* families whose norms scale differently with shape, so a single global η cannot be simultaneously correct for both families and across layers. This is not a lacuna we are obliged to tolerate: the layer-wise LMO framework to which our convergence result reduces (Riabinin et al. 2025; Gruntkowska et al. 2025) already carries per-layer norms $\|\cdot\|_{(\ell)}$, smoothness constants L_ℓ and radii η_ℓ; it is only the *experiments* that have hitherto fixed that radius at a constant. What follows instantiates it. The specific rule is this paper's own; the criterion behind it is borrowed (it is the average-case form of the spectral scaling condition of the maximal-update literature (Yang et al. 2021; Yang, Simon, and Bernstein 2023; Large et al. 2024)), and its LMO endpoint reproduces the aspect factor Muon already uses in practice (Jordan et al. 2024), which is the external check we rely on.

The two families. For a parameter reshaped to $\mathbf{X} \in \mathbb{R}^{m\times n}$ (m = fan-out, n = fan-in, $r = \operatorname{rank} = \min(m, n)$ generically), the step matrices are

$$\mathbf{s} = \mathbf{U}\mathbf{V}^\top \ \ (\text{LMO}), \qquad \mathbf{s} \in \{\pm 1\}^{m\times n} \ \ (\text{SIGN}). \tag{33}$$

The LMO family comprises Muon, MuonUSign, EF21-MuonUSign, EF21-MuonSign and EF21-SignMuon; the SIGN family comprises SignMuon, MuonSign and SignSGD. Both have *exactly* known Frobenius norms: $\|\mathbf{U}\mathbf{V}^\top\|_F^2 = \operatorname{tr}(\mathbf{V}\mathbf{U}^\top\mathbf{U}\mathbf{V}^\top) = \operatorname{tr}(\mathbf{V}^\top\mathbf{V}) = r$, so $\|\mathbf{s}\|_F = \sqrt{r}$; and a ± 1 matrix has $\|\mathbf{s}\|_F = \sqrt{mn}$. (EF21-SignMuon steps along the error-feedback estimator $\mathbf{d}_t^{\text{est}}$ of $\operatorname{polar}(\tilde{\mathbf{M}}_t)$ rather than along the oracle output itself, so for it $\|\mathbf{s}\|_F = \sqrt{r}$ holds only in the limit; we assign it to the LMO family on that basis.)

The criterion. Define the *RMS gain* of $\mathbf{A} \in \mathbb{R}^{m\times n}$, i.e. how much it amplifies a generic input in root-mean-square terms, with $\operatorname{rms}(\mathbf{v}) := \|\mathbf{v}\|/\sqrt{\dim}$:

$$\gamma(\mathbf{A})^2 := \frac{\mathbb{E}_{\mathbf{u}}\big[\operatorname{rms}(\mathbf{A}\mathbf{u})^2\big]}{\mathbb{E}_{\mathbf{u}}[\operatorname{rms}(\mathbf{u})^2]}, \qquad \mathbf{u} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_n). \tag{34}$$

Since $\mathbb{E}\|\mathbf{A}\mathbf{u}\|^2 = \operatorname{tr}(\mathbf{A}^\top\mathbf{A}) = \|\mathbf{A}\|_F^2$ and $\mathbb{E}\|\mathbf{u}\|^2 = n$, (34) evaluates in closed form:

$$\gamma(\mathbf{A}) = \frac{\|\mathbf{A}\|_F}{\sqrt{m}}. \tag{35}$$

No random-matrix theory and no distributional fitting enter; the single modelling assumption is that $\mathbf{u}$ is isotropic *and independent of* $\mathbf{A}$, which we return to below. Controlling a layer update's RMS-to-RMS effect is precisely the desideratum of the spectral scaling condition (Yang, Simon, and Bernstein 2023) and of the modular norm (Large et al. 2024); (34) is its average-case (isotropic-input) version, and when the independence fails, that is, for aligned inputs, where the operator norm governs the gain instead, one recovers the μP endpoint $a = 1$ discussed below.

Every standard fan-in-scaled initialization has *shape-independent* gain. For He normal ($\sigma^2 = 2/n$), $\|\mathbf{X}\|_F = \sigma\sqrt{mn} = \sqrt{2m}$, so $\gamma(\mathbf{X}) = \sqrt{2}$; for PyTorch's default Kaiming-uniform convolution ($\sigma^2 = 1/(3n)$), $\gamma(\mathbf{X}) = 1/\sqrt{3}$. Either way a constant. Requiring the update's gain to be a fixed fraction of the weight's is therefore simply the requirement that *the per-step gain be the same on every layer*, and by (35) that is one formula:

$$\boxed{\eta_\ell = \eta_0\,\lambda_\ell, \qquad \lambda_\ell = \frac{\sqrt{m}}{\|\mathbf{s}\|_F}} \tag{36}$$

which gives $\gamma(\eta_\ell \mathbf{s}) = \eta_0$ exactly, for every shape and both families, so that η_0 *is* the per-step RMS gain. One caveat: the rule is derived from $\|\mathbf{U}\mathbf{V}^\top\|_F = \sqrt{r}$, which holds for the exact oracle. The five-step Newton–Schulz iteration we actually run returns a step 6–22% shorter than $\sqrt{\min(m,n)}$, so for the LMO-terminated methods the realized gain is flat only up to that factor, and η_0 is the per-step RMS gain of the exact step rather than of the executed one. Substituting the two Frobenius norms of (33),

$$\lambda_\ell^{\text{LMO}} = \sqrt{\max\bigl(1, \tfrac{m}{n}\bigr)}, \qquad \lambda_\ell^{\text{SIGN}} = \frac{1}{\sqrt{n}} = \frac{1}{\sqrt{\text{fan-in}}}. \tag{37}$$

Why we believe (37)**.** The first expression is exactly the aspect-ratio factor $\sqrt{\max(1, m/n)}$ present in the reference Muon implementation (Jordan et al. 2024), which was introduced as a practical heuristic. The unit-gain criterion *derives* it, and also explains why Muon's step size is known to transfer across widths: its step has $\gamma = \eta$ independently of n, so no fan-in correction is needed. The second expression is the counterpart the sign family has never been given. Only η_0 is tuned, and it is now a shape-free quantity; the shape dependence is determined a priori.

Two consequences are worth recording. First, λ_ℓ is a deterministic function of the layer shape, known to server and clients alike, so per-layer step sizes require *no* communication and leave the one-bit-per-parameter budget intact. Second, on the CIFAR ResNet-18 of our experiments $\lambda_\ell^{\text{SIGN}}$ spans a factor of 13 across layers (from $1/\sqrt{27}$ at the stem to $1/\sqrt{4608}$ in the last stage), so a single global rate is necessarily a compromise: roughly correct for the middle of the network, several-fold too large at the stem and too small at the end. The LMO family is exempt from this, which is one reason full-precision Muon is easier to tune than its sign-compressed variants.

The one open exponent. Writing $\lambda_\ell^{\text{SIGN}} = n^{-a}$, the unit-gain rule is $a = \frac{1}{2}$. Identity (35) requires the input to be independent of $\mathbf{A}$, which is the right model for a *single* step; if the *accumulated* update $\sum_t \eta \mathbf{s}_t$ correlates with the activations, its gain can be as large as $\|\mathbf{A}\|_{\text{op}}\sqrt{n/m} = \Theta(\eta n)$ rather than $\Theta(\eta\sqrt{n})$, giving $a = 1$, the classical μP rule $\eta \propto 1/\text{fan-in}$ for sign-like updates (Yang et al. 2021). The endpoint is thus an empirical question, and one that a direct measurement settles: we track the realized gain $\|\mathbf{X}_t - \mathbf{X}_0\|_F/\sqrt{m}$ over training and fit its growth exponent h in t: $h = \frac{1}{2}$ for incoherent (random-walk) accumulation, $h = 1$ for aligned accumulation. The fit must be taken at a constant step size. Under a decaying schedule the accumulation saturates and the fit reports the schedule instead of the alignment; on our cosine-annealed runs it returns $|\hat{h}| < 0.06$ at $R^2 \leq 0.31$, which is not a shallow power law but the absence of one. At a constant rate over 20 epochs, $\hat{h} = 0.513$ for Muon and 0.516 for SignMuon, both at $R^2 = 1.000$.

Muon serves as the control. For the LMO family (37) and the μP rule prescribe the identical multiplier, so Muon's exponent is the value the diagnostic returns when the rule is not in question; SignMuon's agreement with it to within 0.003 places the sign family's accumulation in the same regime. The measured value is that of the incoherent model, so $a = \frac{1}{2}$, and we adopt it for both families in every network experiment (the synthetic problem of Appendix A.12 has a single matrix parameter, where the multiplier is one constant absorbed into the tuned η). What the exponent fixes is the scaling of the accumulated update in t, hence the shape dependence of λ_ℓ and the transfer of η_0 across widths; it makes no claim about which exponent maximizes accuracy at one fixed width, where a alters only the relative rates across layers and is largely absorbed into η_0.

The placement of weight decay is not arbitrary. The same scale invariance that makes λ_ℓ necessary also dictates *where* an ℓ_2 penalty may be applied. Every step map considered here is positively homogeneous of degree *zero*: $\text{sign}(c\mathbf{M}) = \text{sign}(\mathbf{M})$ and $\text{polar}(c\mathbf{M}) = \text{polar}(\mathbf{M})$ for all $c > 0$. Consequently, folding a decay term into the gradient before the LMO, that is, setting $\widetilde{\mathbf{G}}_t = \mathbf{G}_t + \lambda_{\text{wd}}\mathbf{X}_t$ after the convention of Mishra, Trivedi, and Kumar (2026) and of most sign-method implementations, supplies no contraction whatever: the step length $\eta_t\lambda_\ell\|\mathbf{s}\|_F$ is fixed by (33) and the decay term cannot alter it, so that any shrinkage which occurs is a by-product of *rotating* the direction rather than the geometric decay the penalty is intended to impose. For the sign-terminated steps the cancellation is exact to machine precision; for the LMO-terminated ones it is exact for the polar factor itself, and the few percent by which a Newton–Schulz approximation of it moves is the iteration's error, not shrinkage. The rotation is governed by the ratio $\rho_t = \lambda_{\text{wd}}\|\mathbf{X}_t\|_F/\|\mathbf{G}_t\|_F$, and it is not benign. Since $\|\mathbf{G}_t\|_F$ falls by orders of magnitude over training while $\|\mathbf{X}_t\|_F$ grows, ρ_t drifts from negligible to $\Theta(1)$; more damaging for a comparison, it depends on each method's own momentum scale, so that one nominal λ_{wd} constitutes a different perturbation for each. Decoupled decay,

Rule	LMO	SIGN	Equalizes
global ($a=0$)	1	1	nothing
Muon default	$\sqrt{\max(1, \frac{m}{n})}$	1	LMO gain
unit gain	$\sqrt{\max(1, \frac{m}{n})}$	$n^{-1/2}$	per-step gain, both
μP ($a=1$)	$\sqrt{\max(1, \frac{m}{n})}$	n^{-1}	aligned accumulation
Mishra et al.	$r^{-1/2}$	$(mn)^{-1/2}$	$\|\mathbf{s}\|_F$

Table A8: Per-layer step-size multipliers λ_ℓ. Only η_0 is tuned; λ_ℓ is fixed a priori by the shape. The rules differ only in the SIGN family, and the LMO column of the unit-gain rule coincides with the factor already employed in practice (Jordan et al. 2024), which is our principal evidence that (34) is the correct criterion. The last row is the normalization of Mishra, Trivedi, and Kumar (2026), which equalizes $\|\mathbf{s}\|_F$ rather than the gain.

$\mathbf{X}_{t+1} = (1-\eta_t\lambda_{\mathrm{wd}})\mathbf{X}_t - \eta_t\lambda_\ell\mathbf{s}_t$, is by contrast exactly commensurate with the update under the unit-gain rule: the displacement it contributes has gain $\eta_t\lambda_{\mathrm{wd}}\gamma(\mathbf{X}_t)$ against the step's gain η_t, so their ratio is $\lambda_{\mathrm{wd}}\gamma(\mathbf{X}_t)$, free of η_t, of the layer shape and of the method. We therefore decouple, and use the coupled form only for an ablation. This also explains an otherwise puzzling observation of Mishra, Trivedi, and Kumar (2026): sweeping $\lambda_{\mathrm{wd}} \in \{0, 0.1, 0.2\}$ coupled over 330 CIFAR-10 ResNet-50 configurations, every one of the five Muon and Sign-Muon entries in their top ten (the only entries for which decay was swept at all) selects $\lambda_{\mathrm{wd}} = 0$. At their two nonzero values $\rho_t \gg 1$ throughout training, so the update degenerates towards $-\operatorname{sign}(\mathbf{X}_t)$; the sweep rejects the placement rather than the regularizer.

What the analysis covers. Our theorems are stated for unregularized f, so we report unregularized runs as the primary comparison and weight decay as an ablation; the reference nanoGPT configuration we build on also uses $\lambda_{\mathrm{wd}} = 0$ for every parameter group. Two remarks delimit the gap. First, the *coupled* form is covered verbatim: it is nothing other than running the same method on $f_\lambda = f + \frac{\lambda_{\mathrm{wd}}}{2}\|\mathbf{X}\|_F^2$, so every rate carries over with $L_i \mapsto L_i + \lambda_{\mathrm{wd}} r_i$. The rank factor is not slack in the bound: our smoothness is measured in the nuclear norm against a spectral-norm displacement (Assumption 2), and $\|\lambda_{\mathrm{wd}}\mathbf{Z}\|_* \le \lambda_{\mathrm{wd}} r_i \|\mathbf{Z}\|_{2\to 2}$ is tight at $\mathbf{Z} = \mathbf{I}_{r_i}$; $L \mapsto L + \lambda_{\mathrm{wd}}$ would be the Euclidean statement, and these rates are not Euclidean. The paradox is that this is precisely the variant which does not regularize. Second, the *decoupled* form is not covered by our rates, yet it furnishes something the analysis requires. Since $\|\mathbf{s}_t\|_F$ is a known constant and $\lambda_\ell\|\mathbf{s}_t\|_F = \sqrt{m}$ identically under (36), the triangle inequality gives $\|\mathbf{X}_{t+1}\|_F \le (1-\eta_t\lambda_{\mathrm{wd}})\|\mathbf{X}_t\|_F + \eta_t\sqrt{m}$, and hence, for any $\eta_t\lambda_{\mathrm{wd}} \le 1$,

$$\gamma(\mathbf{X}_t) \;\le\; \max\bigl\{\gamma(\mathbf{X}_0),\ \lambda_{\mathrm{wd}}^{-1}\bigr\} \quad \text{for all } t, \tag{38}$$

a bound on the layer's gain that is uniform in t and *independent of the layer shape*. Norm-constrained updates are what render this possible: for SGD the step length is data-dependent and no such a priori bound exists. Since layer-wise LMO analyses assume smoothness on a bounded region, (38) is the statement that decoupled decay supplies that region rather than complicating it. A convergence rate for the decoupled variant itself we do not claim.

Sensitivity: what the rule can and cannot affect. Equation (7) is a heuristic, so it is worth being precise about its scope. Each candidate rule, whether one global rate ($\lambda_\ell = 1$), unit gain ($\lambda_\ell = 1/\sqrt{\text{fan-in}}$) or μP ($\lambda_\ell = 1/\text{fan-in}$), shifts the *selected* η_0 by roughly the multiplier it prescribes, which is its purpose; what would signify is whether it shifts the *ordering* of the methods. Two considerations bound the exposure. The LMO family is unaffected: unit gain and μP prescribe the identical multiplier for it, so five of the eight methods, including every one that carries a guarantee, cannot move. And within the SIGN family every method is tuned and reported under the *same* rule, so a wrong exponent rescales all three alike rather than favouring one. What we have not done is re-tune the SIGN family from scratch under the two competing conventions and confirm the ordering directly; that ablation remains outstanding. Federated CNN2 is the natural setting for it, its three matrix parameters spanning a factor of 7.8 in $\lambda_\ell^{\text{SIGN}}$ with no single shape dominating the parameter count, so that an incorrect exponent has the widest scope to manifest itself. We accordingly record the sign-family ordering as resting on the rule to that extent.

Comparison with concurrent work. Mishra, Trivedi, and Kumar (2026) analyse the normalized update $\mathbf{D}_t = \bar{\mathbf{S}}_t/\sqrt{mn}$, justified by $\|\mathbf{D}_t\|_{\mathrm{op}} \le \|\mathbf{D}_t\|_F = 1$ under their spectral-norm smoothness assumption, and remark that updating with $\bar{\mathbf{S}}_t$ directly is equivalent after absorbing $\sqrt{mn}$ into η_t. That equivalence holds for a single matrix but not across layers of differing shape, and their algorithm applies no shape factor, so their experiments use a single global rate as well. The substitution is also loose in a shape-dependent way: $\mathbf{s}$ has rank at most $\min(m,n)$, so $\|\mathbf{s}\|_F \le \sqrt{\min(m,n)}\,\|\mathbf{s}\|_{\mathrm{op}}$ and the substitution of $\sqrt{mn}$ for the operator norm is loose by up to $\sqrt{\min(m,n)}$. That bound itself grows with depth, from $\sqrt{27} \approx 5.2$ at the stem of a ResNet-18 to $\sqrt{512} \approx 22.6$ in the last stage. The induced spectral trust region therefore drifts across the network instead of remaining flat. Table A8 sets the three rules side by side.

Algorithm A1: MuonLMO

Input: tensor $\mathbf{Y}$; Newton-Schulz coefficients $a = 3.4445$, $b = 4.7750$, $c = 2.0315$; iteration count $\mathrm{ns\_steps}$
Output: Muon LMO direction $\mathbf{D} \approx \mathbf{U}\mathbf{V}^\top$ of $\mathbf{Y}$

$\mathbf{Y} \leftarrow \mathrm{ReshapeTo2D}(\mathbf{Y})$ ▷ Flatten tensor to matrix $m \times n$
$\mathbf{Y} \leftarrow \mathbf{Y}/\|\mathbf{Y}\|_F$ ▷ Normalize (optional)
for $k = 1$ to $\mathrm{ns\_steps}$ **do** ▷ 5th-order Newton-Schulz orthogonalization
  $\mathbf{A} \leftarrow \mathbf{Y}\mathbf{Y}^\top$
  $\mathbf{Y} \leftarrow a\,\mathbf{Y} - b\,\mathbf{A}\mathbf{Y} + c\,\mathbf{A}^2\,\mathbf{Y}$
end for
$\mathbf{D} \leftarrow \mathrm{ReshapeToOriginal}(\mathbf{Y})$ ▷ Reshape back to the original tensor shape
return $\mathbf{D}$

A.18 Algorithms

Federated protocol. In the federated version of SignMuon, the server acts as an aggregation node, while the main optimization procedure is performed locally on the clients. At the beginning of communication round t, each client j receives a copy of the current global model and evaluates one stochastic gradient $\mathbf{G}_t^{(j)} = \nabla f_j(\mathbf{X}_{t-1}; \xi_t^{(j)})$ at it. Clients take *no* local parameter steps, so one round is one server step and no client-drift term arises. The released runs accumulate three mini-batches of 64 at fixed weights to save activation memory; since the BatchNorm statistics are frozen (Appendix A.11) the loss is separable across samples, so that average is identically a gradient at batch 192, and we report it as such.

In accordance with the general SignA framework (5), each client maintains its own local momentum buffer $\mathbf{M}_t^{(j)}$. After computing the gradient, the client updates the momentum estimate, applies the LMO (Newton–Schulz orthogonalization, Algorithm A1), and performs element-wise sign compression:

$$\mathbf{M}_t^{(j)} = \mu\mathbf{M}_{t-1}^{(j)} + (1-\mu)\,\mathbf{G}_t^{(j)}, \qquad \mathbf{D}_t^{(j)} = -A\big(\mathbf{M}_t^{(j)}\big), \qquad \mathbf{s}_t^{(j)} = \mathrm{sign}\big(\mathbf{D}_t^{(j)}\big),$$

for $t = 1, \ldots, \mathrm{rounds}$, the gradient being evaluated at $\mathbf{X}_{t-1}$. The momentum is the exponential-moving-average form of (5), matching Algorithms A8–A9 and the framework of Appendix A.10. Each client thus transmits one bit per matrix parameter component. On the server, the received signs are aggregated by majority vote, $\mathbf{s}_t^{\mathrm{agg}} = \mathrm{sign}\big(\sum_{j=1}^N \mathbf{s}_t^{(j)}\big)$, which is ±1 in each component: client messages are ±1 valued by the convention of Section 4, so at an odd client count the vote cannot tie (at an even count a tie is broken by a fair coin). Since momentum has already been applied at the clients, the server updates the matrix parameters directly, $\mathbf{X}_t = \mathbf{X}_{t-1} - \eta_t \mathbf{s}_t^{\mathrm{agg}}$. The final classification layer is exempt from the SignMuon rule and optimized with AdamW.

Federated error feedback. Although federated SignMuon exhibits strong empirical stability, sign-based compression inevitably introduces bias into the descent direction. To compensate for the quantization error and restore convergence guarantees, the error-feedback mechanism is adapted from EF21-Muon (Gruntkowska et al. 2025). The LMO orthogonalization is moved from the clients to the server: each client evaluates the difference between its local momentum $\mathbf{M}_t^{(j)}$ and its previous gradient estimate $\mathbf{g}_{t-1}^{(j)}$, compresses it via scaled sign, and sends only $(\mathbf{s}_t^{(j)}, \alpha_t^{(j)})$. The server aggregates these, updates the global gradient estimator $\mathbf{g}_t = \frac{1}{N}\sum_j \mathbf{g}_t^{(j)}$, and applies the Muon LMO (Algorithm A9). The transmission of one additional scalar $\alpha_t^{(j)}$ per layer constitutes negligible overhead, retaining the uplink at ≈ 1 bit per parameter, with the full model broadcast on the downlink. All federated variants (the baseline majority-vote SignMuon and the error-feedback EF21-MuonUSign and EF21-MuonSign) thus preserve the geometric benefits of Muon-style matrix orthogonalization while maintaining an extremely low communication budget.

All six federated methods are instances of just two templates, separated by *where the Muon LMO is evaluated*. When the sign acts *after* the LMO (the SignMuon family), each client must orthogonalize locally, so the LMO runs on the worker and the client transmits a compressed *direction* (Algorithm A8). When the sign acts *before* the LMO (the MuonUSign/MuonSign family), the client transmits a compressed *gradient*, and the server reconstructs it and applies a single LMO (Algorithm A9). Within each template, a method is fixed by its uplink compressor $\mathcal{C}^{\uparrow} \in \{\mathrm{sign}, \mathrm{EF21}\}$ and downlink compressor $\mathcal{C}^{\downarrow} \in \{\mathrm{exact}, \mathrm{sign}, \mathrm{EF21\text{-}P}\}$; Table A9 lists the six instantiations. As in the centralized setting, both templates are applied per matrix parameter, while vector parameters and the final classification layer are optimized with AdamW.

Algorithm A2: SignMuon

Input: Initial model $\mathbf{X}_0$, momentum coefficient μ, learning rate η_t
Output: Updated model $\mathbf{X}$

$\mathbf{M}_0 \leftarrow 0$
for $t = 1$ to T **do**
  $\mathbf{G}_t \leftarrow \nabla f(\mathbf{X}_{t-1}; \xi_t)$ ▷ Stochastic gradient
  $\mathbf{M}_t \leftarrow \mu \mathbf{M}_{t-1} + (1-\mu)\, \mathbf{G}_t$ ▷ Momentum accumulation
  $\tilde{\mathbf{M}}_t = \begin{cases} \mathbf{M}_t, & \text{(default)}, \\ (1-\mu)\, \mathbf{G}_t + \mu \mathbf{M}_t, & \text{(Nesterov)} \end{cases}$
  $\mathbf{D}_t \leftarrow \mathrm{MuonLMO}(\tilde{\mathbf{M}}_t)$
  $\mathbf{s}_t^{\uparrow} \leftarrow \mathrm{sign}(\mathbf{D}_t)$ ▷ Uplink sign compression
  $\mathbf{X}_t \leftarrow \mathbf{X}_{t-1} - \eta_t \mathbf{s}_t^{\uparrow}$ ▷ Update parameters
end for

Algorithm A3: EF21-SignMuon

Input: Initial model $\mathbf{X}_0$, momentum coefficient μ, learning rate η_t
Output: Updated model $\mathbf{X}$

$\mathbf{M}_0 \leftarrow 0, \quad \mathbf{d}_0^{\mathrm{est}} \leftarrow 0$
for $t = 1$ to T **do**
  $\mathbf{G}_t \leftarrow \nabla f(\mathbf{X}_{t-1}; \xi_t)$ ▷ Stochastic gradient
  $\mathbf{M}_t \leftarrow \mu \mathbf{M}_{t-1} + (1-\mu)\, \mathbf{G}_t$ ▷ Momentum accumulation (EMA)
  $\tilde{\mathbf{M}}_t = \begin{cases} \mathbf{M}_t, & \text{(default)}, \\ (1-\mu)\, \mathbf{G}_t + \mu \mathbf{M}_t, & \text{(Nesterov)} \end{cases}$
  $\mathbf{D}_t \leftarrow \mathrm{MuonLMO}(\tilde{\mathbf{M}}_t)$
  $\Delta_t^{\uparrow} \leftarrow \mathbf{D}_t - \mathbf{d}_{t-1}^{\mathrm{est}}$ ▷ Uplink residual (LMO direction)
  $\alpha_t^{\uparrow} \leftarrow \mathrm{mean}(|\Delta_t^{\uparrow}|)$
  $\mathbf{d}_t^{\mathrm{est}} \leftarrow \mathbf{d}_{t-1}^{\mathrm{est}} + \alpha_t^{\uparrow}\, \mathrm{sign}(\Delta_t^{\uparrow})$ ▷ Uplink EF21
  $\mathbf{X}_t \leftarrow \mathbf{X}_{t-1} - \eta_t \mathbf{d}_t^{\mathrm{est}}$ ▷ Update parameters
end for

Algorithm A4: MuonUSign

Input: Initial model $\mathbf{X}_0$, momentum coefficient μ, learning rate η_t
Output: Updated model $\mathbf{X}$

$\mathbf{M}_0 \leftarrow 0$
for $t = 1$ to T **do**
  $\mathbf{G}_t \leftarrow \nabla f(\mathbf{X}_{t-1}; \xi_t)$ ▷ Stochastic gradient
  $\mathbf{M}_t \leftarrow \mu \mathbf{M}_{t-1} + (1-\mu)\, \mathbf{G}_t$ ▷ Momentum accumulation
  $\tilde{\mathbf{M}}_t = \begin{cases} \mathbf{M}_t, & \text{(default)}, \\ (1-\mu)\, \mathbf{G}_t + \mu \mathbf{M}_t, & \text{(Nesterov)} \end{cases}$
  $\mathbf{s}_t^{\uparrow} \leftarrow \mathrm{sign}(\tilde{\mathbf{M}}_t)$ ▷ Uplink sign compression
  $\mathbf{D}_t \leftarrow \mathrm{MuonLMO}(\mathbf{s}_t^{\uparrow})$
  $\mathbf{X}_t \leftarrow \mathbf{X}_{t-1} - \eta_t \mathbf{D}_t$ ▷ Update parameters
end for

Algorithm A5: MuonSign

Input: Initial model $\mathbf{X}_0$, momentum coefficient μ, learning rate η_t
Output: Updated model $\mathbf{X}$

1. $\mathbf{M}_0 \leftarrow 0$
2. **for** $t = 1$ to T **do**
3. $\quad \mathbf{G}_t \leftarrow \nabla f(\mathbf{X}_{t-1}; \xi_t)$ ▷ Stochastic gradient
4. $\quad \mathbf{M}_t \leftarrow \mu \mathbf{M}_{t-1} + (1-\mu)\,\mathbf{G}_t$ ▷ Momentum accumulation
5. $\quad \tilde{\mathbf{M}}_t = \begin{cases} \mathbf{M}_t, & \text{(default)}, \\ (1-\mu)\,\mathbf{G}_t + \mu \mathbf{M}_t, & \text{(Nesterov)} \end{cases}$
6. $\quad \mathbf{s}_t^{\uparrow} \leftarrow \mathrm{sign}(\tilde{\mathbf{M}}_t)$ ▷ Uplink sign compression
7. $\quad \mathbf{D}_t \leftarrow \mathrm{MuonLMO}(\mathbf{s}_t^{\uparrow})$
8. $\quad \mathbf{s}_t^{\downarrow} \leftarrow \mathrm{sign}(\mathbf{D}_t)$ ▷ Downlink sign compression
9. $\quad \mathbf{X}_t \leftarrow \mathbf{X}_{t-1} - \eta_t \mathbf{s}_t^{\downarrow}$ ▷ Update parameters
10. **end for**

Algorithm A6: EF21-MuonUSign

Input: Initial model $\mathbf{X}_0$, momentum coefficient μ, learning rate η_t
Output: Updated model $\mathbf{X}$

1. $\mathbf{M}_0 \leftarrow 0, \quad \mathbf{g}_0^{\mathrm{est}} \leftarrow 0$
2. **for** $t = 1$ to T **do**
3. $\quad \mathbf{G}_t \leftarrow \nabla f(\mathbf{X}_{t-1}; \xi_t)$ ▷ Stochastic gradient
4. $\quad \mathbf{M}_t \leftarrow \mu \mathbf{M}_{t-1} + (1-\mu)\,\mathbf{G}_t$ ▷ Momentum accumulation
5. $\quad \tilde{\mathbf{M}}_t = \begin{cases} \mathbf{M}_t, & \text{(default)}, \\ (1-\mu)\,\mathbf{G}_t + \mu \mathbf{M}_t, & \text{(Nesterov)} \end{cases}$
6. $\quad \Delta_t^{\uparrow} \leftarrow \tilde{\mathbf{M}}_t - \mathbf{g}_{t-1}^{\mathrm{est}}$ ▷ Uplink residual
7. $\quad \alpha_t^{\uparrow} \leftarrow \mathrm{mean}(|\Delta_t^{\uparrow}|)$
8. $\quad \mathbf{g}_t^{\mathrm{est}} \leftarrow \mathbf{g}_{t-1}^{\mathrm{est}} + \alpha_t^{\uparrow}\,\mathrm{sign}(\Delta_t^{\uparrow})$ ▷ Uplink EF21
9. $\quad \mathbf{D}_t \leftarrow \mathrm{MuonLMO}(\mathbf{g}_t^{\mathrm{est}})$
10. $\quad \mathbf{X}_t \leftarrow \mathbf{X}_{t-1} - \eta_t \mathbf{D}_t$ ▷ Update parameters
11. **end for**

Algorithm A7: EF21-MuonSign

Input: Initial model $\mathbf{X}_0 = \mathbf{W}_0$, momentum coefficient μ, learning rate η_t
Output: Updated model $\mathbf{X}$

1. $\mathbf{M}_0 \leftarrow 0, \quad \mathbf{g}_0^{\mathrm{est}} \leftarrow 0$
2. **for** $t = 1$ to T **do**
3. $\quad \mathbf{G}_t \leftarrow \nabla f(\mathbf{W}_{t-1}; \xi_t)$ ▷ At the broadcast model
4. $\quad \mathbf{M}_t \leftarrow \mu \mathbf{M}_{t-1} + (1-\mu)\,\mathbf{G}_t$ ▷ Momentum accumulation
5. $\quad \tilde{\mathbf{M}}_t = \begin{cases} \mathbf{M}_t, & \text{(default)}, \\ (1-\mu)\,\mathbf{G}_t + \mu \mathbf{M}_t, & \text{(Nesterov)} \end{cases}$
6. $\quad \Delta_t^{\uparrow} \leftarrow \tilde{\mathbf{M}}_t - \mathbf{g}_{t-1}^{\mathrm{est}}$ ▷ Uplink residual
7. $\quad \alpha_t^{\uparrow} \leftarrow \mathrm{mean}(|\Delta_t^{\uparrow}|)$
8. $\quad \mathbf{g}_t^{\mathrm{est}} \leftarrow \mathbf{g}_{t-1}^{\mathrm{est}} + \alpha_t^{\uparrow}\,\mathrm{sign}(\Delta_t^{\uparrow})$ ▷ Uplink EF21
9. $\quad \mathbf{D}_t \leftarrow \mathrm{MuonLMO}(\mathbf{g}_t^{\mathrm{est}})$
10. $\quad \mathbf{X}_t \leftarrow \mathbf{X}_{t-1} - \eta_t \mathbf{D}_t$ ▷ Update parameters (server)
11. $\quad \Delta_t^{\downarrow} \leftarrow \mathbf{X}_t - \mathbf{W}_{t-1}$ ▷ Downlink residual
12. $\quad \alpha_t^{\downarrow} \leftarrow \mathrm{mean}(|\Delta_t^{\downarrow}|)$
13. $\quad \mathbf{W}_t \leftarrow \mathbf{W}_{t-1} + \alpha_t^{\downarrow}\,\mathrm{sign}(\Delta_t^{\downarrow})$ ▷ Downlink EF21-P
14. **end for**

Method	**LMO**	**Uplink $\mathcal{C}^{\uparrow}$**	**Downlink $\mathcal{C}^{\downarrow}$**
SignMuon	worker	sign / MV	exact
EF21-SignMuon	worker	EF21	exact
MuonUSign	server	sign / MV	exact
MuonSign	server	sign / MV	sign
EF21-MuonUSign	server	EF21	exact
EF21-MuonSign	server	EF21	EF21-P

Table A9: The six federated methods as instantiations of the two templates: Algorithm A8 (worker-side LMO; rows 1–2) and Algorithm A9 (server-side LMO; rows 3–6). Each method is fixed by the LMO location and the uplink/downlink compressors. “MV”: majority vote; “EF21-P”: primal (model-side) error feedback; “exact”: full-precision broadcast.

Algorithm A8: Federated SignMuon / EF21-SignMuon (worker-side LMO)

Input: initial model $\mathbf{X}_0$; clients N; rounds T; learning rate η_t; momentum μ; uplink compressor $\mathcal{C}^{\uparrow} \in \{\mathrm{sign}, \mathrm{EF21}\}$ (Table A9)
Output: global model $\mathbf{X}_T$

$\mathbf{M}_0^{(j)} \leftarrow 0$, $\mathbf{d}_0^{(j)} \leftarrow 0$ for all j; $\quad \mathbf{d}_0 \leftarrow 0$
for $t = 1$ to T **do**
  broadcast $\mathbf{X}_{t-1}$ to all clients ▷ exact downlink
  for $j = 1$ to N **in parallel do** ▷ client j
    $\mathbf{G}_t^{(j)} \leftarrow \nabla f_j(\mathbf{X}_{t-1}; \xi_t^{(j)})$
    $\mathbf{M}_t^{(j)} \leftarrow \mu \mathbf{M}_{t-1}^{(j)} + (1-\mu)\, \mathbf{G}_t^{(j)}$
    $\tilde{\mathbf{M}}_t^{(j)} \leftarrow \mathbf{M}_t^{(j)}$ ▷ or $(1-\mu)\mathbf{G}_t^{(j)} + \mu \mathbf{M}_t^{(j)}$ (Nesterov)
    $\mathbf{D}_t^{(j)} \leftarrow \mathrm{MuonLMO}\big(\tilde{\mathbf{M}}_t^{(j)}\big)$ ▷ LMO on the client
    if $\mathcal{C}^{\uparrow} = \mathrm{EF21}$ **then** ▷ EF21-SignMuon
      $\Delta_t^{(j)} \leftarrow \mathbf{D}_t^{(j)} - \mathbf{d}_{t-1}^{(j)}; \quad \alpha_t^{(j)} \leftarrow \mathrm{mean}(|\Delta_t^{(j)}|)$
      $\mathbf{d}_t^{(j)} \leftarrow \mathbf{d}_{t-1}^{(j)} + \alpha_t^{(j)}\, \mathrm{sign}(\Delta_t^{(j)})$
      send $\big(\mathrm{sign}(\Delta_t^{(j)}),\, \alpha_t^{(j)}\big)$
    else ▷ SignMuon
      send $\mathbf{s}_t^{(j)} \leftarrow \mathrm{sign}(\mathbf{D}_t^{(j)})$
    end if
  end for
  on the server:
  if $\mathcal{C}^{\uparrow} = \mathrm{EF21}$ **then**
    $\mathbf{d}_t \leftarrow \mathbf{d}_{t-1} + \frac{1}{N}\sum_j \alpha_t^{(j)}\, \mathrm{sign}(\Delta_t^{(j)})$ ▷ aggregate direction
    $\mathbf{X}_t \leftarrow \mathbf{X}_{t-1} - \eta_t\, \mathbf{d}_t$
  else
    $\hat{\mathbf{s}}_t \leftarrow \mathrm{sign}\big(\sum_j \mathbf{s}_t^{(j)}\big)$ ▷ majority vote
    $\mathbf{X}_t \leftarrow \mathbf{X}_{t-1} - \eta_t\, \hat{\mathbf{s}}_t$
  end if
end for

Algorithm A9: Federated MuonUSign / MuonSign / EF21-MuonUSign / EF21-MuonSign (server-side LMO)

Input: initial model $\mathbf{X}_0$; clients N; rounds T; learning rate η_t; momentum μ; compressors $\mathcal{C}^{\uparrow} \in \{\text{sign}, \text{EF21}\}$, $\mathcal{C}^{\downarrow} \in \{\text{exact}, \text{sign}, \text{EF21-P}\}$ (Table A9)
Output: global model $\mathbf{X}_T$

$\mathbf{M}_0^{(j)} \leftarrow 0$, $\mathbf{g}_0^{(j)} \leftarrow 0$ for all j; $\quad \mathbf{g}_0 \leftarrow 0$
broadcast $\mathbf{X}_0$ *once*; every client holds the broadcast model $\mathbf{W}_0 \leftarrow \mathbf{X}_0$, refreshed below from the downlink message alone ($\mathbf{W}_t = \mathbf{X}_t$ unless $\mathcal{C}^{\downarrow} = \text{EF21-P}$)
for $t = 1$ to T **do**
  for $j = 1$ to N **in parallel do** ▷ client j, holding $\mathbf{W}_{t-1}$
    $\mathbf{G}_t^{(j)} \leftarrow \nabla f_j(\mathbf{W}_{t-1};\, \xi_t^{(j)})$
    $\mathbf{M}_t^{(j)} \leftarrow \mu \mathbf{M}_{t-1}^{(j)} + (1-\mu)\, \mathbf{G}_t^{(j)}$
    $\tilde{\mathbf{M}}_t^{(j)} \leftarrow \mathbf{M}_t^{(j)}$ ▷ or $(1-\mu)\mathbf{G}_t^{(j)} + \mu \mathbf{M}_t^{(j)}$ (Nesterov)
    if $\mathcal{C}^{\uparrow} = \text{EF21}$ **then** ▷ EF21-MuonUSign / EF21-MuonSign
      $\Delta_t^{(j)} \leftarrow \tilde{\mathbf{M}}_t^{(j)} - \mathbf{g}_{t-1}^{(j)}; \quad \alpha_t^{(j)} \leftarrow \operatorname{mean}(|\Delta_t^{(j)}|)$
      $\mathbf{g}_t^{(j)} \leftarrow \mathbf{g}_{t-1}^{(j)} + \alpha_t^{(j)} \operatorname{sign}(\Delta_t^{(j)})$
      send $\big(\operatorname{sign}(\Delta_t^{(j)}),\, \alpha_t^{(j)}\big)$
    else ▷ MuonUSign / MuonSign
      send $\mathbf{s}_t^{(j)} \leftarrow \operatorname{sign}\big(\tilde{\mathbf{M}}_t^{(j)}\big)$
    end if
  end for
  on the server:
  if $\mathcal{C}^{\uparrow} = \text{EF21}$ **then**
    $\mathbf{g}_t \leftarrow \mathbf{g}_{t-1} + \frac{1}{N} \sum_j \alpha_t^{(j)} \operatorname{sign}(\Delta_t^{(j)})$ ▷ reconstruct gradient
  else
    $\mathbf{g}_t \leftarrow \operatorname{sign}\big(\sum_j \mathbf{s}_t^{(j)}\big)$ ▷ majority vote
  end if
  $\mathbf{D}_t \leftarrow \operatorname{MuonLMO}(\mathbf{g}_t)$ ▷ single LMO on the server
  if $\mathcal{C}^{\downarrow} = \text{sign}$ **then** ▷ MuonSign: 1 bit/param down
    $\mathbf{X}_t \leftarrow \mathbf{X}_{t-1} - \eta_t \operatorname{sign}(\mathbf{D}_t)$; **broadcast** $\operatorname{sign}(\mathbf{D}_t)$
    $\mathbf{W}_t \leftarrow \mathbf{W}_{t-1} - \eta_t \operatorname{sign}(\mathbf{D}_t)$ ▷ $= \mathbf{X}_t$; one model
  else if $\mathcal{C}^{\downarrow} = \text{EF21-P}$ **then** ▷ EF21-MuonSign: 1 bit/param down
    $\mathbf{X}_t \leftarrow \mathbf{X}_{t-1} - \eta_t \mathbf{D}_t$ ▷ server model
    $\Delta_t^{\downarrow} \leftarrow \mathbf{X}_t - \mathbf{W}_{t-1}; \quad \alpha_t^{\downarrow} \leftarrow \operatorname{mean}(|\Delta_t^{\downarrow}|)$
    $\mathbf{W}_t \leftarrow \mathbf{W}_{t-1} + \alpha_t^{\downarrow} \operatorname{sign}(\Delta_t^{\downarrow})$; **broadcast** $\big(\operatorname{sign}(\Delta_t^{\downarrow}), \alpha_t^{\downarrow}\big)$ ▷ clients apply the same refresh
  else ▷ exact downlink: MuonUSign / EF21-MuonUSign
    $\mathbf{X}_t \leftarrow \mathbf{X}_{t-1} - \eta_t \mathbf{D}_t$; **broadcast** $\mathbf{X}_t$; $\quad \mathbf{W}_t \leftarrow \mathbf{X}_t$
  end if
end for

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